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# Dynamic prediction of product competitive position: A multisource data-driven competitive analysis framework from a multi-competitor perspective

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ABSTRACT

Competitive analysis is critical in enhancing product quality and facilitating strategic adaptation, with user-generated content (UGC) offering new avenues to improve product competitiveness. Existing UGC-based SWOT analysis methods have made progress in minimizing subjective bias. However, they still lack in analysis scope, enhancement of dynamic forecasting, and improvement of factor prioritization. Given the rapid changes in the market environment, companies urgently require innovative approaches to secure a competitive advantage. From a multi-competitor perspective, this study introduces a multisource data-driven dynamic SWOT (D-SWOT) analysis framework, which broadens the analysis to include all potential market competitors, aiming to dynamically predict the focal product's position in a specific competitive environment and evaluate factor priorities. Initially, competitors are identified from multisource data through a competitor identification algorithm. Subsequently, attributes are identified, their importance is assessed using the LDA model, and their performance is evaluated through the bidirectional long-short-term memory model. This study aims to comprehend market dynamics and formulate effective strategies by employing a grey prediction model to forecast future attribute trends. Based on the core principles of SWOT analysis, rules for determining factors have been clarified, and a D-SWOT matrix has been constructed to predict changes in the product's market position. Finally, integrating attribute performance differences with their relative importance, a competitive priority index apt for competitive analysis has been developed to guide product enhancements and strategic realignments. Through a case study involving tablet products, the framework's substantial benefits in terms of competitive analysis have been validated.

## 1. Introduction

In intensely competitive markets, businesses are compelled to perpetually enhance the quality of their products and develop distinctive competitive advantages to differentiate themselves from their competitors (Zheng et al., 2024). Competitive analysis is crucial for tackling these threats and spotting market opportunities (Bergen and Peteraf, 2002). Its core function is to aid companies in discerning customer perceptions of differences between their products and competitors and devise competitive strategies (Lin et al., 2023). SWOT analysis is widely acknowledged as a pivotal instrument for evaluating competitive positions. It facilitates the precise determination of a product's strategic placement in the market by assessing its internal strengths and weaknesses as well as external opportunities and threats (Tu et al., 2021). The simplicity of the method and the clarity of the results have led to its extensive use across different sectors (Büyüközkan et al., 2021; D'Adamo et al., 2020; Lee et al., 2021; Stylianou et al., 2024). It is considered a highly reliable and crucial tool for analyzing complex decision-making problems (Wang et al., 2020).

Although traditional SWOT analysis is widely utilized, its limitations have been extensively criticized. First, traditional SWOT analysis heavily relies on subjective judgments, leading to overly abstract categorizations of SWOT factors, which lack clarity and precision (Panagiotou, 2003). Second, traditional SWOT analysis lacks mechanisms for dynamic tracking and real-time updating (Coman and Ronen, 2009), hindering its ability to effectively forecast future market trends or adapt to environmental changes, thus impacting timely strategic adjustments by businesses. Third, traditional SWOT analysis fails to

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0969-6989/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

adequately consider factor priorities, resulting in a lack of practical guidance for decision-makers in resource allocation. Finally, the traditional approach fails to consider the market environment fully, making it challenging to accurately identify the market's overall competitive situation and clearly define the specific directions for product improvement.

As information technology advances, more and more consumers share their product experiences online, generating a vast amount of user-generated content (UGC). UGC originates directly from users' authentic feedback, containing extensive user opinions (Cong et al., 2023), and comprehensive competitor information is provided (Kim et al., 2025). Compared with traditional data collection methods for competitive analysis, such as surveys and interviews, UGC offers advantages like easier accessibility, reduced time consumption, and lower cost. Furthermore, advancements in data mining technologies have made the acquisition of multisource data more practical, enabling the collection of data from online reviews, Questions and Answers (Q&A), and expert-generated content (Liu et al., 2021; Qian et al., 2024). This approach mitigates the biases that might arise from relying solely on single-source data (Kou et al., 2021). Existing studies have shown that SWOT analysis based on UGC is extremely valuable for businesses to understand the market environment and identify their competitive positions (Cheng et al., 2021; Wu et al., 2024a).

However, while existing studies have attempted to utilize UGC to address the shortcomings of SWOT analysis and enhance its persuasiveness, there remain deficiencies that have motivated this research:

First, the current research mainly focuses on a limited number of competitors (Lin et al., 2023; Srinivas and Rajendran, 2019), which does not adequately capture the complexity of the competitive landscape. Second, the existing studies on dynamics predominantly rely on retrospective analysis using historical data (Cheng et al., 2021; Liu et al., 2024; Wu et al., 2024b). This approach shows significant limitations in responding to fast market changes and dynamic feedback, potentially weakening decision-making capabilities when developing long-term strategies. Additionally, the regional boundaries of SWOT factors remain vague in existing studies (Cheng et al., 2021; Lin et al., 2023; Tu et al., 2021), and there is a need to develop more accurate methods for delineating these boundaries. Finally, although previous studies have made progress in addressing the issue of factor ranking (Liu et al., 2024; Wu et al., 2024b), they have not paid sufficient attention to performance differences, thereby failing to reflect the specificity of competitive strategies adequately.

From the identified research gaps, this study introduces a multisource data-driven dynamic SWOT (D-SWOT) analysis framework based on a multi-competitor perspective. Specifically, the study broadens the analytical scope by gathering multisource data. It treats the focal product (FP)and all its competitors as a unified competitive market known as the focal product market (FPM). This approach enables a more comprehensive analysis of a product's relative position within a given competitive environment. Subsequently, a grey prediction model is incorporated in place of traditional retrospective methods, providing a more forward-looking perspective in the analysis. The $\eta$-translation line is further introduced in this study, offering a clear standard for the regional division of SWOT factors. Lastly, integrating importance with performance differences to determine factor priorities offers a scientific foundation for implementing product optimization schemes and competitive strategies.

The rest of this study is organized as follows: Section 2 presents the relevant literature. Section 3 describes the proposed methodology. Section 4 presents a case study using tablet data from Zhihu.com and JD.com. Section 5 analyzes the results and presents a discussion on comparative analysis. Finally, Section 6 presents the conclusions and implications of this study.

## 2. Related works

### 2.1. Application of NLP in UGC

UGC offers valuable information for companies developing strategic plans as the digital economy grows. Analyzing this information effectively to understand the underlying drivers of geopolitical, economic, social, technological, and structural transformations has become a crucial topic (Gaponenko, 2022). In the UGC domain, natural language processing (NLP) technology plays a critical role, primarily in feature extraction and sentiment analysis, and researchers extensively use it in industries including hotels (Zhao et al., 2023), smart devices (Liu et al., 2023; Zhang et al., 2018), tourist destinations (Gour et al., 2021), automobiles (Wang et al., 2023), green product consumption (Wang et al., 2024), and healthcare (Zhai et al., 2022).

The multidimensionality of user opinions may reflect several factors, which are key in purchasing decisions and assist companies in distinguishing their products in the market. Consequently, the latent dirichlet allocation (LDA) model is widely used as an unsupervised learning algorithm (Wang et al., 2018; Yang et al., 2024), designed to identify and describe the most common and potential attributes in unstructured datasets (Huang et al., 2022). Even when some attributes are not apparent in documents, the Bayesian inference process can identify them through iterative refinement, condensing large volumes of information into a few key attributes (Pavlinek and Podgorelec, 2017). The LDA model is favored in text analysis for its interpretability and scalability and has been extensively applied in extracting attributes and keywords (Qian et al., 2023). It also plays a critical role in competitive analysis. For example, Zhao et al. (2023) used LDA to identify key attributes impacting hotel competitiveness, Wang et al. (2018) applied it to derive positive and negative attributes of two competitors, Qian et al. (2024) expanded it to extract expert opinions and comparisons of different product attributes from professionally generated content, and Guo et al. (2022) introduced the brand joint LDA model to analyze both general and specific aspects across multiple brands. This model has shown impressive capabilities and extensive application potential in the multidimensional opinion analysis of UGC, offering robust support for competitive analysis, product optimization, and strategic planning for companies.

Sentiment analysis allows companies to compare user opinions between products, helping them optimize their competitive strategies. UGC-based sentiment analysis methods mainly include dictionary-based methods (Zhang et al., 2023) and machine learning-based methods (Liu et al., 2024). Machine learning is typically more accurate than dictionary-based methods when dealing with complex sentences, particularly in large-scale datasets (Liu et al., 2024). Recently, several studies have applied machine learning-based sentiment analysis to competitive analysis (Bi et al., 2019; Chen et al., 2025; Wu et al., 2024a; Zheng et al., 2024). For instance, the studies by Chen et al. (2025) and Wu et al. (2024a) reveal the powerful capabilities of machine learning in sentiment analysis, which significantly enhances companies' understanding and grasp of market competition through deep exploration and analysis of user sentiment feedback on social media. In addition, machine learning-based sentiment analysis techniques are also applied in fields such as satisfaction analysis (Kumar et al., 2024) and product recommendation (Wang et al., 2024). These applications powerfully demonstrate the effectiveness of text mining techniques in extracting valuable business opportunities from UGC.

### 2.2. Prediction methods

The successful formulation of competitive strategies requires the early prediction of product attribute trends to determine improvement directions preemptively (Wang et al., 2024). Numerous scholars have developed practical forecasting methods leveraging historical data to mitigate market uncertainty risks and help businesses sustain their

competitive advantage in highly competitive markets.

In terms of forecasting techniques, several researchers have suggested employing machine learning methods such as support vector machines, random forest, long short-term memory (LSTM), and graph neural networks (Lu et al., 2021; Ping et al., 2021; Zaghloul et al., 2024; Zhang et al., 2024). However, these methods rely on large datasets and struggle to adapt to short-term and small-sample scenarios. Conversely, fuzzy time series, fuzzy rough set time series, and methods combining the kano model with the grey-markov model (Jiang et al., 2017; Kwong et al., 2016; Shieh and Wu, 2009; Yakubu and Kwong, 2021) exhibit lower data requirements, making them more suitable for data-constrained environments. Nevertheless, these approaches generally rely on questionnaire data or expert surveys and impose stringent requirements on data distribution, limiting their effectiveness in predicting user demand when sample sizes are inadequate.

In recent research, the grey prediction model (Wang et al., 2024), the markov chain (Alexandrovskiy and Trundova, 2022), and the auto-regressive and moving average (ARIMA) (Wu et al., 2024) have been applied to demand forecasting in UGC data. The grey prediction model exhibits remarkable predictive capability, given its high tolerance for small sample sizes and incomplete information. For instance, Wang et al. (2024) employed the grey prediction model to forecast product attribute priorities. Zhang et al. (2023) applied the grey-markov model to forecast attribute attention and satisfaction. Liu and Ma (2024) employed the multivariate grey prediction model to predict user demand. These studies underscore the exceptional capability of the grey prediction model in processing UGC data, suggesting its promising potential for future applications.

### 2.3. SWOT analysis

SWOT analysis serves as a powerful tool for evaluating both the internal and external environments of a company, offering an internal perspective to effectively assess its strengths and weaknesses (Büyüközkan and Ilıcak, 2019). Typically, a SWOT analysis consists of four key components: internal strengths and weaknesses, external opportunities and threats. These components help companies diagnose their situation while providing robust strategic planning support. The SWOT analysis matrix enables companies to identify their position relative to competitors and create effective adaptive strategies (Cheng et al., 2021).

Although researchers widely use SWOT analysis, existing research highlights several limitations of the traditional method, which could seriously impede a company's ongoing competitiveness. Firstly, traditional SWOT analysis often lacks clear and quantifiable evaluation criteria(Cheng et al., 2021), making the results vulnerable to the subjective judgment of analysts, thus diminishing both the quality of the analysis and the validity of the outcomes. To address this, researchers have proposed quantitative methods to enhance the objectivity and reliability of the analysis. For example, Tu et al. (2021) proposed a method for determining factor categories based on average performance values, Lin et al. (2023) used a quantile mapping method to divide data into four intervals based on the 25 %, 50 %, and 75 % quartiles, Cheng et al. (2021) combined change mining and association rule mining techniques for the classification of SWOT factors, and Wu et al. (2024b) introduced a method that considers performance differences. Although these methods have enhanced the accuracy of SWOT analysis, ambiguity in the regionalization of factors persists, and further improvement and refinement are required.

Furthermore, traditional SWOT analysis tends to be static, usually reflecting a company's internal and external conditions at a specific time (Coman and Ronen, 2009), making it difficult to perceive reality from a dynamic evolutionary perspective (Vlados, 2019). User opinions are shaped by continuously changing economic, social, environmental, and technological factors and may exhibit varying demands at different product lifecycle stages (Zhang et al., 2024). In response to these changes, companies must identify and incorporate dynamic user opinions early in the product design (Sha et al., 2024). Although retrospective methods have been used in some studies to describe the dynamic changes in SWOT quantitatively (Cheng et al., 2021; Wu et al., 2024b), these studies are dependent on historical data, meaning that their results merely reflect past or present market conditions. Consequently, they fail to account for the emerging threats and potential opportunities that products may encounter in a rapidly evolving market environment, constraining their effectiveness in guiding improvement strategies. In today's business environment, retrospective analysis is no longer sufficient to comprehensively assess a firm's actual competitive position, which in turn compromises the foresight and adaptability of strategic decision-making.

Third, the prioritization of factors has not been effectively evaluated in traditional SWOT analysis (Büyüközkan and Ilıcak, 2019; Namugenyi et al., 2019). Although improvement priorities have been determined in some studies using importance indicators derived from surveys and expert interviews, combined with methods such as multiple linear regression (Phadermrod et al., 2019) or the analytic hierarchy process (Lee et al., 2021; Wang et al., 2022), the subjectivity inherent in these questionnaires and expert-based data has constrained the practical applicability of SWOT analysis. In this regard, Wu et al. (2024b) determined improvement priorities by leveraging large-scale UGC data and computing attribute importance, thereby mitigating the interference of subjective factors to some extent. However, the ranking of factors based on text or word frequency merely indicates the frequency of an attribute and its keywords in the text rather than its actual impact on decision-making or strategic importance.

Fourth, the analytical scope of traditional SWOT analysis is also subject to certain limitations. SWOT analysis methods usually determine the scope of analysis in three ways: first, the qualitative analysis method, where competitors are identified based on expert opinions, relevant research, or industry reports, as seen in the works of Lee et al. (2021) and Büyüközkan et al. (2021). This method depends on expert judgment or past industry experience, with an emphasis on subjective reasoning and analysis. The second method is the quantitative analysis approach, in which representative competitors are identified through data collected from surveys, as shown in the research by Wang et al. (2022)and Phadermrod et al. (2019). Finally, the data mining analysis method is employed, in which the primary competitors in the market are identified using a word co-occurrence matrix from online reviews, as seen in the study by Lin et al. (2023). While these methods assist in identifying competitors to some extent, they limit the number of competitors considered and fail to reflect the complexity of the competitive market adequately.

Table 1 provides a summary of recent studies on SWOT analysis. UGC-based research has increasingly attracted academic attention, as its extensive data offers strong support for SWOT analysis. Based on these previous findings, this study extends the depth and breadth of research. This study adopts a multi-competitor perspective and introduces a multisource data-driven D-SWOT analysis framework. This framework is designed for specific competitive environments and relies entirely on quantitative data, effectively mitigating the subjectivity limitations of traditional SWOT analysis. Furthermore, this framework resolves the limitations of existing SWOT methods in dynamic analysis and priority determination, thereby facilitating the optimization and refinement of SWOT factors.

## 3. Methodology

From a multi-competitor perspective, this study introduces a multisource data-driven competitive analysis framework—D-SWOT. The framework is structured in four stages: First, competitor identification: After the FP is determined, online reviews and Q&A related to the product are collected, and a competitor identification algorithm is developed to match the competitor set, followed by preprocessing of the

**Table 1**
Summary of recent literature on SWOT analysis.

| Author | Data source | Competitor | Priority | Dynamic analysis |
|---|---|---|---|---|
| Wang et al. (2022) | Questionnaire | / | Importance | / |
| Farrokhnia et al. (2024) | Literature | / | / | / |
| Lee et al. (2021) | Expert opinions, related research, and industry reports | Several competitors | Importance | / |
| Cheng et al. (2021) | UGC | Several competitors | / | Retrospective |
| Phadermrod et al. (2019) | Questionnaire | Single competitor | Importance | / |
| Büyüközkan et al. (2021) | Expert opinions, related research, and industry reports | / | Importance | / |
| Wu et al. (2024b) | UGC | Several competitors | Importance | Retrospective |
| Lin et al. (2023) | UGC | Several competitors | / | / |
| Srinivas and Rajendran (2019) | UGC | Several competitors | / | / |
| Tu et al. (2021) | UGC | Single competitor | / | / |
| Liu et al. (2024) | UGC | Single competitor | Importance& Performance | Retrospective |
| Our study | UGC | All competitors | Importance& Performance differences | Predictive |

product reviews. Second, attribute importance calculation and performance evaluation: In this stage, the LDA model extracts product attributes and calculates their importance. At the same time, sentiment analysis is performed using a bidirectional long-short-term memory (Bi-LSTM) model to assess the performance of each attribute. Thirdly, dynamic forecasting utilizes the grey prediction model: During this phase, the grey prediction model takes the importance and attribute performances of FP and FPM as inputs to project future attribute trends. Finally, D-SWOT and strategic prioritization: Using the forecasts from the grey prediction model, we construct a D-SWOT matrix that translates attributes into SWOT factors, revealing the time-evolution trends of SWOT factors for FP. Moreover, this study introduces the competitive priority index (CPI), which integrates attribute performance differences and importance to establish the priority ranking for SWOT factor improvements. The framework proposed by this research is depicted in Fig. 1.

### 3.1. Competitor identification and data preprocessing

Before buying a product, users tend to compare products with similar prices (Ma et al., 2024) or similar functions (Qian et al., 2023). They often browse reviews and related forums for similar products and raise questions like "Which is better, Vivo Pad 2 or Xiaomi Pad 6 Pro?" or "How should I choose between Huawei Mate Pad Air and Xiaomi Pad 6 Pro?" In these discussions, users frequently share their experiences. Even when the question concerns a single product, such as "How is the Xiaomi Pad 6 Pro?", the answers may still include comparisons and recommendations for other products. Therefore, these Q&A interactions need to be analyzed to gather complete competitor product information.

To accurately identify competitor products (CPs) from a large volume of Q&A, a web scraping tool collected all available product names from the tablet product pages on JD.com, creating a product name set, *P_set*. A standardization function was developed to address differences in letter case, Chinese vocabulary, and spacing. For each product $p$ in the set, the *normalize_product_name* function was applied to standardize the

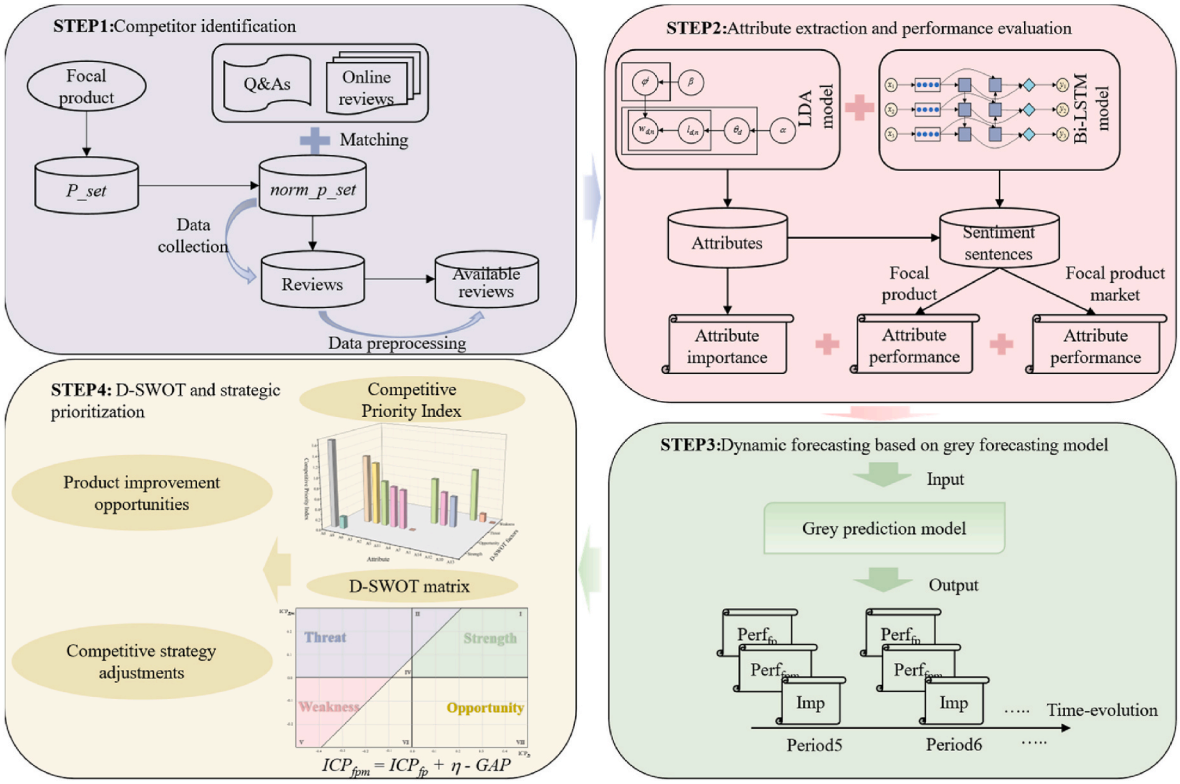



**Fig. 1.** The framework proposed in this research.

product name, producing the standardized name *norm_p*. For example, "小米pad6 Pro," "xiaomi pad6 Pro," and "Xiaomipad6Pro" are all standardized to "Xiaomi Pad 6 Pro." Finally, a function named *is_competitor* was defined to determine whether product *p* is a CP. It identifies products that appear alongside the FP as CPs and outputs them in the *comP_set*. The detailed process is shown in Table 2.

After the competitor product set *comP_set* is established, all relevant reviews from CPs are collected. The data preprocessing involves two steps: data cleaning and text processing. In the data cleaning phase, duplicate reviews and irrelevant information are removed. A custom dictionary containing industry-specific terms and jargon was constructed during the text-processing phase to improve the model's accuracy. Word segmentation is done using the Chinese word segmentation tool Jieba, and all non-Chinese characters are removed. A synonym dictionary was also established to improve semantic understanding, and a stopword dictionary was introduced to reduce noise and irrelevant terms.

### 3.2. Attribute importance calculation and performance evaluation

#### 3.2.1. Attribute extraction and importance calculation

This study applies LDA to extract attributes. LDA is a generative probabilistic model applied to discrete datasets, such as text corpora. It aims to efficiently process extensive collections by identifying concise representations of the set members and preserving statistical relationships crucial for classification, novelty detection, summarization, similarity assessment, and relevance assessment. The central idea of LDA is that documents are viewed as random mixtures of latent attributes, with each attribute characterized by a distribution of words (Blei et al., 2003).

Suppose a review $d$ is expressed as a sequence of words $w = (w_1, w_2, \ldots, w_n)$, where $w_n$ represents the $n$-th word in the sequence. All Q&As from FP and CPs are collected as the corpus D.

1. Choose $N \sim Possion\ (\xi)$.
2. Choose $\theta \sim Dir\ (\alpha)$.
3. For each of the $N$ words $w_n$:
   (a) Choose an attribute $\varphi^i \sim$ Multinomial $(\theta)$.
   (b) Choose a word $w_n$ from $p(w_n|\varphi^i, \beta)$ a multinomial probability conditioned on the attribute $\varphi^i$.

LDA is a hierarchical Bayesian model with three levels, as depicted in Fig. 2. The outermost layer is the review layer, the middle layer is the word layer, and the top layer is the attribute layer. $\theta_d$ follows a Dirichlet distribution with parameter $\alpha$, representing the attribute distribution for the $d$-th review. $\varphi^i$ follows a Dirichlet distribution with parameter $\beta$, representing the word distribution for the $i$-th attribute (Wang et al., 2024).

**Table 2**
Competitor identification algorithm.

**Algorithm 1**. Identifying competitors from the set of Q&As

**Input:**
*P_set*, the set of all products
*Q&As*, a collection of UGC that includes questions and answers
*P_target*, the product for which competitors are being identified
**Output:**
*comP_set*, a set of products identified as competitors to *P_target*
**Steps:**
1: *comP_set* ← ∅
2: *norm_P_set* ← ∅
3: *norm_P_target* ← normalize_product_name(*P_target*)
4: for p in *P_set* do
  5: *norm_p* ← normalize_product_name(*p*)
  6: *norm_P_set.add(norm_p)*
7: end for 8: for *Q&A_item* in *Q&As* do
  9: for p in *norm_P_set* do
    10: if *is_competitor(p, norm_P_target, Q&A_item)* then
      11: *comP_set.add(p)*
    12: end if
  13: end for
14: end for
15: return *comP_set*

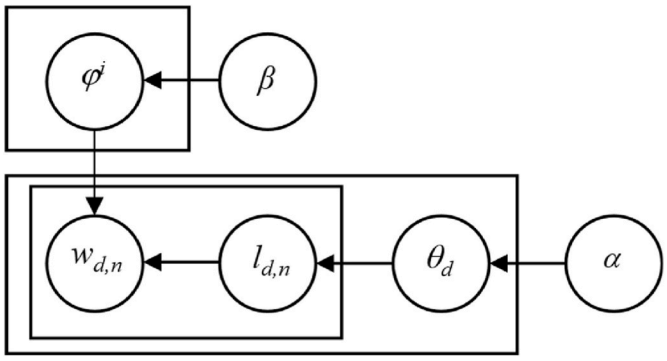



**Fig. 2.** Graphical model for LDA.

The LDA model's performance is assessed using two primary metrics: perplexity and coherence. Perplexity quantifies the model's predictive capability, where lower values indicate higher accuracy and reduced uncertainty. The coherence score evaluates the semantic consistency and interpretability of the attributes identified by the model. A higher coherence score suggests that words within an attribute are more semantically related, resulting in a more precise representation of attributes in the review dataset (Park et al., 2023). Generally, lower perplexity and higher coherence scores reflect better model performance (Guo et al., 2022).

For reviews involving multiple attributes, attribute importance can be defined by the level of user attention it receives, which is typically reflected in the frequency of its appearance in user reviews. Using the review-attribute probability matrix generated by LDA for attribute $A^i$, one can locate reviews associated with this attribute and calculate the total number of its mentions ($M^i$). Attribute importance is then assessed by the ratio of these mentions ($M^i$) to the total number of reviews ($M$) in the corpus, as expressed in Equation (1).

$$\mathrm{Imp}^i = \frac{M^i}{M} \tag{1}$$

#### 3.2.2. Bi-LSTM-based attribute performance evaluation

This study employs a sentiment analysis model to predict sentiment classification tasks, providing a more precise assessment of each attribute's performance. By integrating LDA output with manually labeled data, we segment each review into multiple sentences to better capture sentiment in reviews with multiple attributes, enhancing the model's fine-grained analytical capabilities. The basic architecture of the Bi-LSTM model includes an input layer, embedding layer, bidirectional long short-term memory layer, dropout layer, and output layer. The embedding layer of the Bi-LSTM model employs bidirectional encoder representations from transformers (BERT). This pre-trained bidirectional language representation model utilizes a self-attention mechanism to unify word vector inputs and incorporates bidirectional cross-attention between sentence pairs. This capability enables the Bi-LSTM model to recognize relevant words and fully capture sentence-level contextual meaning (Choudhary and Arora, 2024), allowing the same word to have distinct embedding representations across different sentences (Xiao et al., 2022), thereby more comprehensively capturing text features. This method has been widely applied in NLP (Cho et al., 2024).

The architecture of the Bi-LSTM model is illustrated in Fig. 3. In the preprocessing phase, we use the BERT tokenizer for text segmentation, appending [CLS] to the beginning of the text sequence as a classification token and [SEP] to the end as a separator. The BERT model converts the

text into dense vectors, each corresponding to a token in the input sequence. These vectors are fed into the bidirectional LSTM layer, where the forward LSTM processes the current input vector based on the previous hidden state. At the same time, the backward LSTM incorporates the hidden state from the next step. This setup enables the Bi-LSTM to process forward and backward information (Li et al., 2020). The dropout technique is applied during training to mitigate overfitting by randomly deactivating neurons, enhancing the model's generalization capability. Finally, the output layer, comprising a fully connected layer and a softmax activation function, produces the sentiment classification results: positive (1), neutral (0), and negative (−1).

The attribute performance of FP and FPM was computed to analyze the sentiment distribution of attributes. $sentiment^{i}_{fp,h}$ and $sentiment^{i}_{fpm,h}$ denote the sentiment scores of attribute $A^i$ in sentence $h$ for FP and FPM, respectively. $M^{i}_{fp}$ and $M^{i}_{fpm}$ indicate the number of sentences associated with attribute $A^i$ in FP and FPM, respectively.

$$Perf^{i}_{fp} = \frac{\sum_{h=1}^{M^i} sentiment^{i}_{fp,h}}{M^{i}_{fp}}$$
$$Perf^{i}_{fpm} = \frac{\sum_{h=1}^{M^i} sentiment^{i}_{fpm,h}}{M^{i}_{fpm}} \quad (2)$$

### 3.3. Dynamic forecasting based on grey prediction model

The dynamic changes in the market frequently result in delayed product improvement strategies, making the prediction of attributes especially crucial. Considering the substantial differences in user preferences and behavioral patterns at various product lifecycle stages, this study utilizes the grey prediction model to analyze attributes.

First, the magnitude ratio value test is conducted on the original sequence, which comprises FP performance, FPM performance, and importance across different periods, denoted as $Perf_{fp}$, $Perf_{fpm}$, and $Imp$, respectively. The original sequence is assumed to be $X^{(0)}=\{x^{(0)}(1), x^{(0)}(2), \ldots, x^{(0)}(k), \ldots, x^{(0)}(s)\}$, where periods are represented by $k$, with $k = 1,2, \ldots, s$. The equation for the magnitude ratio value test is given as follows:

$$\lambda_k = \frac{x^{(0)}(k-1)}{x^{(0)}(k)} \quad (3)$$

Here, $k = 2,3, \ldots, s$, If $\lambda_k$ lies within the interval $\left[e^{-\frac{2}{s+1}}, e^{\frac{2}{s+1}}\right]$, the original sequence satisfies the necessary conditions for the grey prediction model. Otherwise, a transformation is required, where an appropriate constant $b$ is selected to keep $\lambda_k$ within a reasonable range, followed by a magnitude ratio value test. The transformation equation is given below:

$$y^{(0)}(k) = x^{(0)}(k) + b \quad (4)$$

Next, the raw data are converted into a first-order cumulative sequence. The newly generated sequence is: $X^{(1)} = \{x^{(1)}(1), x^{(1)}(2), \ldots, x^{(1)}(k), \ldots, x^{(1)}(s)\}$. Then, the first-order differential equation and the grey differential equation for $x^{(1)}$ are established. The first-order differential equation is given as follows:

$$\frac{dx^{(1)}(k)}{dt} + az^{(1)}(k) = u \quad (5)$$

The grey differential equation for the grey prediction model is given as follows:

$$x^{(0)}(k) + az^{(1)}(k) = u \quad (6)$$

$z^{(1)}(k) = \frac{1}{2}\left(x^{(1)}(k) + x^{(1)}(k-1)\right)$, and $Z^{(1)} = \left\{z^{(1)}(2), z^{(1)}(3), \ldots, z^{(1)}(k), \ldots, z^{(1)}(s)\right\}$, where $a$ denotes the development coefficient, and $u$ represents the grey force. The values of $a$ and $u$ are estimated using the least squares method, as given below:

$$\hat{a} = \begin{bmatrix} a \\ u \end{bmatrix} = \left(B^T B\right)^{-1} B^T Y_S \quad (7)$$

$$B = \begin{pmatrix} -z^{(1)}(2)1 \\ -z^{(1)}(3)1 \\ \ldots \\ -z^{(1)}(s)1 \end{pmatrix}, Y_s = \begin{pmatrix} X^{(0)}(2) \\ X^{(0)}(3) \\ \ldots \\ X^{(0)}(s) \end{pmatrix} \quad (8)$$

$$\hat{x}^{(1)}(k+1) = \left(x^{(0)}(1) - \frac{u}{a}\right)e^{-ak} + \frac{u}{a} \quad (9)$$

$$\hat{x}^{(0)}(k+1) = \hat{x}^{(1)}(k+1) - \hat{x}^{(1)}(k), k = 1, 2, \ldots, s-1 \quad (10)$$

Finally, the accuracy of this model must be evaluated. According to Wang et al. (2024), the variance ratio ($C$) and small error probability ($P$) are used as criteria for evaluating model accuracy. The $C$ value assesses the model's fit to historical data. A lower $C$ value indicates smaller prediction errors and higher accuracy. The $P$ value quantifies the proportion of small errors between predicted and actual values. A higher $P$ value indicates greater prediction accuracy. The reference criteria are

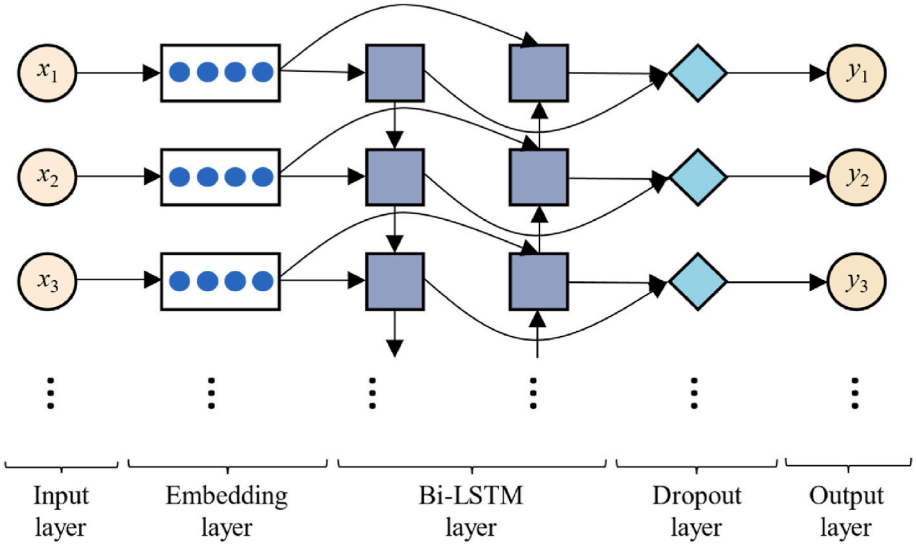



**Fig. 3.** Architecture of the Bi-LSTM model.

summarized in Table 3.

In addition to the above metrics, this study introduces the relative error and magnitude ratio deviation values to analyze the model's residuals further and assess its accuracy. The relative error quantifies model accuracy by computing the relative difference between predicted and actual values. The magnitude ratio deviation assesses whether the rank comparison values in the model deviate from theoretical values, thereby determining its applicability and predictive stability.

### 3.4. Constructing the D-SWOT matrix and competitive strategic prioritization

#### 3.4.1. Construction of the D-SWOT matrix

Following the general steps of SWOT analysis, we first determine the attribute's internal competitive position (ICP) based on the performance (Perf) of FP and FPM, then identify FP's SWOT factors based on the performance differences (denoted as GAP) between FP and FPM.

To be precise, the product's internal average performance (IAP) is first calculated. If $Perf > IAP$, then $ICP > 0$, representing internal strength; if $ICP < 0$, it is classified as an internal weakness. Next, the performance differences between FP and FPM are computed. $GAP > 0$ indicates a positive performance difference; if $GAP < 0$, it represents a negative performance difference between FP and FPM. For a given attribute $A^i$, if $ICP^i_{fp} > 0$, it is considered an internal strength; if $ICP^i_{fpm} > 0$ and $GAP^i > 0$, it is classified as a strength; if $GAP^i < 0$, it is categorized as a threat. If $ICP^i_{fpm} \leq 0$ and $GAP^i > 0$, it is classified as an opportunity; if $GAP^i < 0$, it is considered a threat. Similarly, when $ICP^i_{fp} \leq 0$, it is classified as an internal weakness. If $ICP^i_{fpm} > 0$ and $GAP^i > 0$, it is considered an opportunity; if $GAP^i < 0$, it is categorized as a threat. If $ICP^i_{fpm} \leq 0$ and $GAP^i > 0$, it is regarded as an opportunity, whereas if $GAP^i < 0$, it is classified as a weakness. The detailed rules are presented in Table 4.

Based on the aforementioned classification results, for attribute $A^i$, $ICP_{fp}$ and $ICP_{fpm}$ are defined as the coordinate axes of the two-dimensional D-SWOT matrix. The data points corresponding to $ICP^i_{fp}$ and $ICP^i_{fpm}$ are positioned as points in the SWOT matrix. Define:

$$\eta = IAP_{fp} - IAP_{fpm} \tag{11}$$

Based on these transformations, the $\eta$-translation line for the D-SWOT matrix has been defined as follows:

$$ICP_{fp_m} = ICP_{fp} + \eta - GAP \tag{12}$$

When $GAP = 0$, the $\eta$-translation line of the D-SWOT matrix is given by $ICP_{fpm} = ICP_{fp} + \eta$. When $\eta > 0$, the $\eta$-translation line crosses the first, second, and third quadrants. When $\eta < 0$, it traverses the first, third, and fourth quadrants. If $\eta = 0$, the $\eta$-translation line passes through the origin of the coordinate system. Regions where $GAP > 0$ are located in the lower right of the $\eta$-translation line, corresponding to the D-SWOT factors of strength and opportunity. Conversely, regions where $GAP < 0$ are positioned in the upper left, aligning with weakness and threat.

This study builds on the preceding analysis to conduct an in-depth exploration of how variations in $\eta$ influence the matrix structure and introduces three types of D-SWOT matrix diagrams, as illustrated in Fig. 4. When $\eta = 0$, the SWOT matrix is segmented into six regions: Region I represents strengths, Region IV represents weaknesses, Regions II and III indicate threats, and Regions V and VI denote opportunities (see Fig. 4(b)). When $\eta \neq 0$, the matrix is divided into seven regions. If $\eta > 0$, the newly introduced Region VII is classified as an opportunity (see Fig. 4(a)). Conversely, If $\eta < 0$, Region VII is categorized as a threat (see Fig. 4(c)).

**Table 3**
Reference standards.

| Model | Good | Qualified | Barely | Unqualified |
|---|---|---|---|---|
| *P* value | >0.95 | >0.80 | >0.70 | ≤0.70 |
| *C* value | <0.35 | <0.45 | <0.50 | ≥0.65 |

#### 3.4.2. Determination of competitive priorities

This study integrates the opportunity algorithm into competitive analysis to further refine the prioritization of improvements in D-SWOT factors. Previous research suggests that attributes deemed important but insufficiently satisfied represent innovation opportunities, guiding the future development trajectory of FP. Accordingly, an opportunity algorithm based on result-driven innovation was developed, allowing businesses to identify and prioritize growth opportunities (Jeong et al., 2019).

Inspired by this concept, this study incorporates the impact of performance differences and introduces CPI. Specifically, Equation (13) is designed to assess improvement opportunities by integrating importance with the performance differences between FP and FPM. The attribute's improvement priority increases when $Imp'_t - GAP'_t > 0$. Consequently, a smaller $GAP_t$' value for an attribute corresponds to a higher improvement priority. This method guarantees that resource allocation strategically targets reinforcing key attributes relatively weaker in competition and exhibiting lower performance differences, ultimately improving the product's competitiveness in the overall market. For attribute $A^i$, CPI is computed as follows:

$$\begin{aligned} &CPI_t = Imp'_t + \left(Max\left(Imp'_t - GAP'_t\right), 0\right) \\ &Imp'_t = (Imp_t - Imp_{\min})/(Imp_{\max} - Imp_{\min}); \\ \text{Where, } &GAP'_t = (GAP_t - GAP_{\min})/(GAP_{\max} - GAP_{\min}); \\ &GAP_t = Perf_{t,fp} - Perf_{t,fpm} \end{aligned} \tag{13}$$

Market conditions, consumer demand, technological advances, and the competitive landscape constantly evolve, and a single SWOT analysis may not promptly reflect the dynamic changes in the marketplace. By using predictive models, companies can proactively forecast market trends and combine predicted changes with SWOT analysis to help companies adapt ahead of time. This way, companies can respond more quickly and flexibly to market changes rather than passively dealing with existing problems.

From the above discussion, this study addresses key issues in product strategy planning, competitive positioning, and setting future development priorities by comprehensively applying various data analysis methods. It provides companies with a comprehensive, objective, and forward-looking decision-support framework.

**Table 4**
Rules for determining SWOT factors.

| Focal product | | Focal product market | | *GAP* | SWOT factor |
|---|---|---|---|---|---|
| $ICP_{fp}$ | Internal strength/ weakness | $ICP_{fpm}$ | Internal strength/ weakness | | |
| >0 | S | >0 | S | >0 | S |
| >0 | S | >0 | S | >0 | S |
| >0 | S | >0 | S | <0 | T |
| >0 | S | >0 | S | <0 | T |
| >0 | S | ≤0 | W | >0 | O |
| >0 | S | ≤0 | W | >0 | O |
| >0 | S | ≤0 | W | <0 | T |
| >0 | S | ≤0 | W | <0 | T |
| ≤0 | W | >0 | S | >0 | O |
| ≤0 | W | >0 | S | >0 | O |
| ≤0 | W | >0 | S | <0 | T |
| ≤0 | W | >0 | S | <0 | T |
| ≤0 | W | ≤0 | W | >0 | O |
| ≤0 | W | ≤0 | W | >0 | O |
| ≤0 | W | ≤0 | W | <0 | W |
| ≤0 | W | ≤0 | W | <0 | W |

Note: S stands for Strength, O for Opportunity, T for Threat, and W for Weakness. For Attribute $A^i$, (1) $GAP = Perf^i_{fp} - Perf^i_{fpm}$; (2) $ICP^i_{fp} = Perf^i_{fp} - IAP_{fp}$, $ICP^i_{fpm} = Perf^i_{fpm} - IAP_{fpm}$, where $i = (1,2, \ldots,m)$, $Perf^i_{fp}$ refers to the attribute performance of the FP, while $Perf^i_{fpm}$ refers to the attribute performance of the FPM.

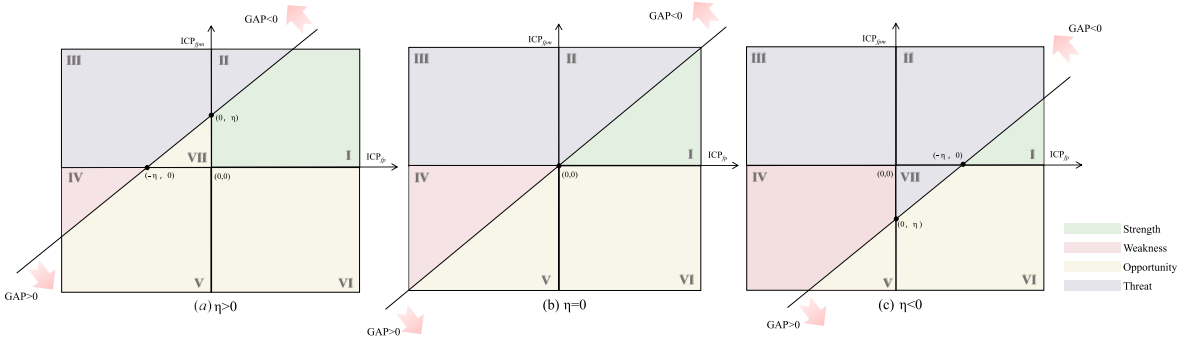



**Fig. 4.** The D-SWOT matrix proposed in this research.

## 4. Case study

The present study proposes a framework and utilizes the tablet Xiaomi Pad 6 Pro as FP to validate its feasibility. The study identified CPs from multiple sources, including the highly influential Chinese e-commerce platform JD.com (https://www.jd.com) and the Chinese Q&A community Zhihu.com (https://www.zhihu.com). The JD.com platform enables users who have previously procured products to submit reviews and facilitates inquiries from prospective buyers. Zhihu.com is a community where users share knowledge, experiences, and insights.

To obtain relevant data, we employed web crawling techniques to extract information on the Xiaomi Pad 6 Pro from JD and Zhihu, yielding 8154 Q&A entries and 10,166 reviews. After further processing, entries with empty content, duplicates, and invalid data were removed, resulting in 5084 Q&A entries and 9837 valid reviews.

### 4.1. Competitor identification results and data preprocessing

Following the process outlined in Section 3.1 to derive the *comP_set*, the related products were ranked by frequency of occurrence, as illustrated in Table 5. Among these, the top four CPs of the Xiaomi Pad 6 Pro are very similar in launch date and price. This finding supports the argument made by Ma et al. (2024) that products with similar launch dates and prices have a clear competitive relationship. Furthermore, five other products were frequently mentioned, suggesting that consumers, when selecting products, not only focus on options with similar prices but also tend to compare those with higher cost-performance ratios or better performance.

Next, 124,483 review entries were collected based on the *comP_set*. These data cover UGC from July 2023 to June 2024, including user IDs, review content, review dates, and product model information. We removed invalid reviews during the data cleaning and kept 102,084 valid reviews, providing a rich LDA analysis corpus. To ensure the accuracy of the analysis, we carried out tokenization, removed non-Chinese characters, replaced synonyms, and eliminated stop words, among other operations. Moreover, considering the significant differences in the number of reviews across different periods, which could introduce outliers and affect the stability of the analysis, we divided the review data into four periods for analysis, each containing three months. Table 6 shows the number of reviews for each period.

### 4.2. Attribute extraction and performance evaluation results

#### 4.2.1. Attribute extraction results

In user reviews, multiple attributes are often mentioned. Each review is segmented into numerous sentences based on commas and periods to enhance the accuracy of model identification. A unique index is constructed by combining the user ID and review date. By incorporating perplexity and coherence metrics, the LDA model determined that the optimal number of attributes ranges from 11 to 17. Following the consolidation of similar keywords, 14 attributes were extracted and subsequently named according to expert judgment. Table 7 displays the 14 attributes and their respective top 5 keywords. Table 8 lists the attribute importance across four periods.

**Table 5**
Products information.

| Product name | Launch date | Launch price | Number of reviews |
|---|---|---|---|
| Xiaomi Pad 6 pro | 2023.4 | 1899–3299 | 9837 |
| Vivo pad 2 | 2023.4 | 2399–3399 | 8813 |
| IQOO pad2 | 2023.5 | 2499–3399 | 9886 |
| Oppo pad 2 | 2023.3 | 2999–3999 | 10,792 |
| Mate pad air | 2023.5 | 2899–3899 | 9937 |
| Xiaomi Pad 6 S pro | 2024.2 | 2699–4499 | 9980 |
| Xiaomi Pad 5 pro | 2021.8 | 2499–3499 | 11,336 |
| Ipad9 | 2021.9 | 2499–4899 | 13,316 |
| Mate pad 11.5 | 2024.5 | 2099–3299 | 10,238 |
| Honor v8 pro | 2022.12 | 2499–3299 | 7949 |

#### 4.2.2. Attribute performance evaluation results

This study divided the labeled dataset into training, validation, and test sets in an 8:1:1 ratio. The model achieved an accuracy of 0.91, a mean precision of 0.92, a mean recall of 0.98, and a mean F1 score of 0.95. The trained Bi-LSTM model demonstrated strong predictive performance. Table 9 summarizes the attribute performance of the Xiaomi Pad 6 Pro and its market trends across different periods.

### 4.3. Dynamic forecasting results

This study employs the grey prediction model to forecast attribute importance and performance, which were extracted in previous steps. The prediction process is illustrated with $A^1$ as an example.

First, the data undergoes the magnitude ratio value test. Table 10 presents the test results $\lambda$ ranging from 0.822 to 1.260, which satisfies the standard within the $\left[e^{-\frac{2}{5}}, e^{\frac{2}{5}}\right]$ range and is appropriate for constructing the grey prediction model.

Next, the attribute importance and performance for the four periods are input into the grey prediction model, yielding the development coefficient $a$ and grey force $u$, as shown in Table 11. The model accuracy test results indicate that the $C$ value of $A^1$ is less than 0.65, within the range [0.0001, 0.5628], and the $P$ value is over 0.70, within the range [0.75, 1], indicating that the model meets the required standards.

In addition, this study uses residual tests to assess the model's fit.

**Table 6**
Number of reviews for each period.

| Period | Period1 | Period2 | Period3 | Period4 |
|---|---|---|---|---|
| Number of effective reviews | 18,558 | 24,908 | 24,050 | 34,568 |

**Table 7**
Attributes and corresponding keywords.

| Abbreviation | Attribute | Keywords |
|---|---|---|
| $A^1$ | Appearance | Aesthetic, Design, Color, Attractive, Material |
| $A^2$ | Size | Portability, Ratio, Weight, Compact, Frame |
| $A^3$ | Display Quality | Image Quality, Pixel Density, Stunning, Binge-watching, Gaming |
| $A^4$ | Quality | Quality Control, Workmanship, Expectation, Body, Metal |
| $A^5$ | Sound | Sound Effect, Sound Quality, Surround, Excellent, Movies |
| $A^6$ | Performance | Processor, Heat, Frame Rate, Speed, Lag |
| $A^7$ | Battery | Battery life, Time, Power Consumption, Charging, Background |
| $A^8$ | Ecosystem | Interconnection, Smartphone, Sync, Convenience, Collaboration |
| $A^9$ | Photography | Photography, Pixel, Photo, Mode, Clarity |
| $A^{10}$ | Express Delivery | Shipping, Order, Speed, Damage, Protection |
| $A^{11}$ | Cost performance | Price, Cost-effective, Discount, Subsidy, Configuration |
| $A^{12}$ | Customer Service | Service Attitude, Price Reduction, Price Guarantee, Return, Vendor |
| $A^{13}$ | Software | Compatibility, Gaming, Optimization, Software, Landscape Mode |
| $A^{14}$ | Memory | Space, Storage, Occupation, Lag, Tasks |

Tables 12–14 present the relative error values and magnitude ratio deviation values of Xiaomi Pad 6 pro's attribute performance ($Perf_{xiaomi}$), its competitive market's attribute performance ($Perf_m$), and importance (Imp). The relative error value remains below 0.2, and the magnitude ratio deviation value is below 0.2, demonstrating that the model fits well and the predictions are valid. Table 15 presents the predicted results for all attributes over the subsequent two periods. Fig. 5 illustrates the actual and forecasted results across all periods.

### 4.4. Construction of the D-SWOT matrix and competitive strategic prioritization results

#### 4.4.1. D-SWOT matrix results

Based on the grey prediction model's forecasted results, the required indicators for constructing the D-SWOT matrix—$ICP_{Xiaomi}$, $ICP_m$, and GAP, as presented in Table 16 and Table 17—were computed. The values of $\eta$ for Period 3 through Period 6 are 0.0508, 0.0855, 0.0914, and 0.1072, respectively. Given that $\eta > 0$, the D-SWOT matrix traverses the first, second, and third quadrants, resulting in the final D-SWOT matrix depicted in Fig. 6.

Ensuring the effectiveness of future decision-making and the precision of strategic planning requires a focus on analyzing the dynamics and variations during the forecasting period. Period 3 and Period 4 represent actual periods, whereas Period 5 and Period 6 correspond to forecast periods. In Period 4, the strengths declined from four to three, the threats dropped from four to two, and the number of opportunities increased. Detailed differences are presented in Table 16.

**Table 8**
The importance of attributes for each period.

| Attribute | Period1 | Period2 | Period3 | Period4 |
|---|---|---|---|---|
| $A^1$ | 0.4285 | 0.3400 | 0.3205 | 0.2995 |
| $A^2$ | 0.4145 | 0.3130 | 0.3390 | 0.4260 |
| $A^3$ | 0.2940 | 0.3860 | 0.4175 | 0.4490 |
| $A^4$ | 0.3580 | 0.3115 | 0.3230 | 0.1695 |
| $A^5$ | 0.4955 | 0.3030 | 0.3225 | 0.3560 |
| $A^6$ | 0.3025 | 0.2835 | 0.3040 | 0.4200 |
| $A^7$ | 0.2845 | 0.3600 | 0.4315 | 0.3710 |
| $A^8$ | 0.3875 | 0.3655 | 0.3570 | 0.4920 |
| $A^9$ | 0.4645 | 0.3500 | 0.3120 | 0.2685 |
| $A^{10}$ | 0.3750 | 0.3570 | 0.2680 | 0.2550 |
| $A^{11}$ | 0.3260 | 0.3735 | 0.3585 | 0.4145 |
| $A^{12}$ | 0.3055 | 0.3780 | 0.4400 | 0.3820 |
| $A^{13}$ | 0.2385 | 0.2015 | 0.2320 | 0.1265 |
| $A^{14}$ | 0.3650 | 0.4225 | 0.3640 | 0.3680 |

In period 5, under the condition that $GAP > 0$, $A^8$ and $A^9$ are identified as strengths, with $ICP_{Xiaomi} > 0$ and $ICP_m > 0$, positioning them in Region I. $A^2$, $A^3$, $A^4$, $A^5$, $A^6$, $A^7$, and $A^{11}$ are categorized as opportunities. Regarding their positions, $A^5$ and $A^6$ exhibit $ICP_{Xiaomi} < 0$ and $ICP_m < 0$, concurrently, situating them in Region V. $A^2$, $A^4$, and $A^{11}$ exhibit $ICP_{Xiaomi} > 0$ but $ICP_m < 0$, placing them in Region VI. $A^3$ and $A^7$ exhibit $ICP_{Xiaomi} < 0$ but $ICP_m > 0$, situating them in Region VII. In the event of $GAP < 0$, $A^1$ and $A^{14}$ exhibit $ICP_{Xiaomi} < 0$ but $ICP_m > 0$, categorizing them as threats in Region III. $A^{10}$, $A^{12}$, and $A^{13}$ exhibit $ICP_{Xiaomi} < 0$ and $ICP_m < 0$ concurrently, classifying them as weaknesses. Regarding factor dynamics, as $A^3$ transitioned from $ICP_{Xiaomi} > 0$ in Period 4 to $ICP_{xiaomi} < 0$, this attribute shifted from a strength to an opportunity. As $A^{10}$ transitioned from $ICP_m > 0$ to $ICP_m < 0$, it was reclassified as a weakness. $A^{14}$, which transitioned from $GAP > 0$ to $GAP < 0$, exhibited weaker market performance and was reclassified from an opportunity to a threat.

In period 6, while attribute performance has changed, the competitive position remains broadly stable. The only notable change is that $A^7$ has transitioned into a threat, reflecting a shift in user sentiment. This shift suggests that the company should closely monitor this attribute and consider reevaluating relevant strategies to mitigate potential negative impacts on the product's competitive position.

#### 4.4.2. Competitive strategic prioritization results

The CPI value identifies the key attributes within each SWOT factor. Fig. 7 illustrates the evolution of competitive priorities over the forecast period, facilitating the identification of product attributes critical for future market competition.

In period 5, $A^8$ was identified as a key strength attribute, representing the core competitive advantage. Sustaining this competitive advantage requires continuous investment in research and development, optimization, and innovation to secure the product's leading market position. $A^3$ and $A^6$ are key opportunity attributes. Despite their $ICP_{Xiaomi} < 0$, both still outperform FPM. Ongoing improvements can widen the gap between these attributes and other CPs, reinforcing a competitive advantage. $A^{14}$ is a key threat attribute, while $A^{12}$ is a key weakness attribute. Both should be prioritized for improvement. Resources should be allocated correctly to prevent these attributes from becoming obstacles to the product's market performance. Non-key attributes should retain their current resource allocation or be scaled down appropriately to avoid resource wastage, provided key attributes receive sufficient investment. This differentiated resource allocation strategy enhances core competitiveness by preventing excessive investment in attributes with limited market value and optimizing resource utilization.

In contrast to the observations made in period 5, there are notable shifts in period 6. In detail, $A^6$ assumes a higher priority ranking than $A^3$, $A^{14}$'s priority experiences a decline, and A7 transitions from a secondary to a primary threat, becoming a pivotal attribute in this context. Consequently, businesses should be cognizant of these potential changes, address the status and variations of these key attributes, assess their possible impact on product or brand image and competitive positioning, and develop appropriate risk mitigation strategies. The specific changes in the priority of key attributes are outlined in Table 18.

A comprehensive analysis of key attributes, attribute keywords, and associated reviews reveals that users prioritize $A^8$'s interconnectivity, data synchronization, collaboration with mobile phones, and use as an extended screen. For $A^6$, the main concerns are processor type, tablet heating, response speed, and smoothness during daily use. In the case of $A^3$, users emphasize image quality and viewing comfort when watching movies. $A^{14}$'s core concerns include storage space and lag issues. For $A^7$, users pay particular attention to battery life and charging speed. $A^{12}$ users prioritize the merchant's service attitude, price reductions, and return guarantees. These user insights are critical for enhancing competitiveness. By fostering a forward-thinking approach, anticipating

**Table 9**
The performance of attributes for each period.

| Attribute | Period1 | | Period2 | | Period3 | | Period4 | |
|---|---|---|---|---|---|---|---|---|
| | $Perf_{xiaomi}$ | $Perf_m$ | $Perf_{xiaomi}$ | $Perf_m$ | $Perf_{xiaomi}$ | $Perf_m$ | $Perf_{xiaomi}$ | $Perf_m$ |
| $A^1$ | 0.5511 | 0.5654 | 0.6299 | 0.6027 | 0.6232 | 0.7328 | 0.5356 | 0.6348 |
| $A^2$ | 0.6505 | 0.7222 | 0.7343 | 0.6246 | 0.7917 | 0.6382 | 0.8761 | 0.5177 |
| $A^3$ | 0.7364 | 0.5889 | 0.7250 | 0.6521 | 0.6207 | 0.6620 | 0.6721 | 0.5829 |
| $A^4$ | 0.7564 | 0.6296 | 0.7765 | 0.6318 | 0.8725 | 0.6305 | 0.8543 | 0.5487 |
| $A^5$ | 0.6660 | 0.5908 | 0.7292 | 0.6724 | 0.7464 | 0.6441 | 0.6187 | 0.5645 |
| $A^6$ | 0.5560 | 0.5665 | 0.6382 | 0.6040 | 0.5734 | 0.5007 | 0.6183 | 0.4573 |
| $A^7$ | 0.6876 | 0.6222 | 0.7235 | 0.6270 | 0.7294 | 0.6182 | 0.6344 | 0.6172 |
| $A^8$ | 0.7950 | 0.7116 | 0.7820 | 0.7126 | 0.7644 | 0.7218 | 0.8019 | 0.7411 |
| $A^9$ | 0.7565 | 0.6057 | 0.6765 | 0.6057 | 0.8780 | 0.6215 | 0.8672 | 0.6590 |
| $A^{10}$ | 0.7022 | 0.7698 | 0.6379 | 0.7048 | 0.6160 | 0.6609 | 0.5569 | 0.5872 |
| $A^{11}$ | 0.8657 | 0.6820 | 0.8149 | 0.6466 | 0.8587 | 0.6538 | 0.7410 | 0.5503 |
| $A^{12}$ | 0.6045 | 0.3921 | 0.5864 | 0.4701 | 0.4515 | 0.4625 | 0.4417 | 0.4519 |
| $A^{13}$ | 0.7030 | 0.7511 | 0.6456 | 0.7299 | 0.4663 | 0.6232 | 0.4227 | 0.5356 |
| $A^{14}$ | 0.7659 | 0.6677 | 0.7365 | 0.7104 | 0.5881 | 0.6977 | 0.6435 | 0.6385 |

**Table 10**
Results of the ratio test for $A^1$.

| Period | $Perf^1_{xiaomi}$ | | $Perf^1_m$ | | $Imp$ | | Range $[e^{-2/5}, e^{2/5}]$ |
|---|---|---|---|---|---|---|---|
| | Raw value | $\lambda$ | Raw value | $\lambda$ | Raw value | $\lambda$ | |
| 1 | 0.551 | – | 0.565 | – | 0.428 | – | Yes |
| 2 | 0.630 | 0.875 | 0.603 | 0.938 | 0.340 | 1.260 | Yes |
| 3 | 0.623 | 1.011 | 0.733 | 0.822 | 0.321 | 1.061 | Yes |
| 4 | 0.536 | 1.164 | 0.635 | 1.154 | 0.299 | 1.070 | Yes |

**Table 11**
Model construction results for $A^1$.

| | Development coefficient $a$ | Grey force $u$ | $C$ value | $P$ value |
|---|---|---|---|---|
| $Perf^1_{xiaomi}$ | 0.0773 | 0.7103 | 0.1691 | 1 |
| $Perf^1_m$ | −0.0231 | 0.6212 | 0.5628 | 0.75 |
| $Imp$ | 0.0632 | 0.3783 | 0.0001 | 1 |

**Table 12**
Residual test results for $Perf_{xiaomi}$ of $A^1$.

| Period | Raw value | Forecasting value | Residual | Relative error | Magnitude ratio deviation |
|---|---|---|---|---|---|
| 1 | 0.551 | 0.551 | 0.000 | 0.000 | – |
| 2 | 0.630 | 0.643 | −0.013 | 0.020 | 0.190 |
| 3 | 0.623 | 0.595 | 0.028 | 0.046 | 0.065 |
| 4 | 0.536 | 0.551 | −0.015 | 0.028 | −0.077 |

**Table 13**
Residual test results for $Perf_m$ of $A^1$.

| Period | Raw value | Forecasting value | Residual | Relative error | Magnitude ratio deviation |
|---|---|---|---|---|---|
| 1 | 0.565 | 0.565 | 0.000 | 0.000 | – |
| 2 | 0.603 | 0.642 | −0.039 | 0.065 | 0.040 |
| 3 | 0.733 | 0.657 | 0.076 | 0.104 | 0.158 |
| 4 | 0.635 | 0.672 | −0.037 | 0.059 | −0.181 |

**Table 14**
Residual test results for $Imp$ of $A^1$.

| Period | Raw value | Forecasting value | Residual | Relative error | Magnitude ratio deviation |
|---|---|---|---|---|---|
| 1 | 0.428 | 0.428 | 0.000 | 0.000 | – |
| 2 | 0.340 | 0.340 | 0.000 | 0.001 | −0.183 |
| 3 | 0.321 | 0.319 | 0.001 | 0.003 | 0.004 |
| 4 | 0.299 | 0.300 | 0.000 | 0.001 | −0.005 |

**Table 15**
Forecasting results for all attributes over the next two periods.

| Attribute | Period5 | | | Period6 | | |
|---|---|---|---|---|---|---|
| | $Perf_{xiaomi}$ | $Perf_m$ | $Imp$ | $Perf_{xiaomi}$ | $Perf_m$ | $Imp$ |
| $A^1$ | 0.5096 | 0.6877 | 0.2820 | 0.4718 | 0.7038 | 0.264 |
| $A^2$ | 0.9534 | 0.4975 | 0.4900 | 1.0421 | 0.4562 | 0.576 |
| $A^3$ | 0.6194 | 0.5675 | 0.4840 | 0.5946 | 0.5380 | 0.522 |
| $A^4$ | 0.9132 | 0.5267 | 0.1350 | 0.9558 | 0.4924 | 0.075 |
| $A^5$ | 0.5978 | 0.5277 | 0.3810 | 0.5538 | 0.4848 | 0.409 |
| $A^6$ | 0.5897 | 0.3876 | 0.5030 | 0.5798 | 0.3358 | 0.622 |
| $A^7$ | 0.6131 | 0.6110 | 0.3980 | 0.5759 | 0.6062 | 0.403 |
| $A^8$ | 0.8031 | 0.7543 | 0.5560 | 0.8136 | 0.7693 | 0.656 |
| $A^9$ | 1.0067 | 0.6842 | 0.2370 | 1.1271 | 0.7140 | 0.208 |
| $A^{10}$ | 0.5276 | 0.5425 | 0.2010 | 0.4937 | 0.4960 | 0.168 |
| $A^{11}$ | 0.7359 | 0.5290 | 0.4260 | 0.7040 | 0.4904 | 0.451 |
| $A^{12}$ | 0.3598 | 0.4436 | 0.4040 | 0.3088 | 0.4350 | 0.406 |
| $A^{13}$ | 0.3182 | 0.4576 | 0.1150 | 0.2536 | 0.3919 | 0.081 |
| $A^{14}$ | 0.5637 | 0.6140 | 0.3320 | 0.5231 | 0.5828 | 0.3090 |
| IAP | **0.6508** | **0.5593** | 0.3531 | **0.6427** | **0.5355** | 0.3679 |

market shifts, and developing contingency strategies, companies can effectively adapt to external dynamics and thrive in a highly competitive landscape.

## 5. Discussion

### 5.1. Comparison of sentiment analysis models

Sentiment analysis constitutes a crucial part of this study, designed to capture user perceptions precisely. To this end, the widely adopted Bi-LSTM model was utilized, and for reliability verification, LSTM, Convolutional Neural Network (CNN), and BERT models were concurrently trained on the same corpus. The accuracy of the four models was assessed by comparing them under identical parameter settings. Additionally, to minimize the influence of true negatives on the results, the precision, recall, and F1 scores of each model were calculated, as shown in Fig. 8. The results demonstrate that, given identical configurations and parameter settings, the Bi-LSTM model surpasses the other three models in performance.

### 5.2. Comparison of prediction methods

In the dynamic prediction phase, this study investigates three alternative methods: the grey-markov model, the ARIMA model, and the fractional grey model. As per Wu et al. (2018), when the fractional order r = 1, the fractional grey model reduces to the conventional grey prediction model. Consequently, this study employs r = 0.3, r = 0.5, and r = 0.8 to explore how changes in fractional order affect prediction

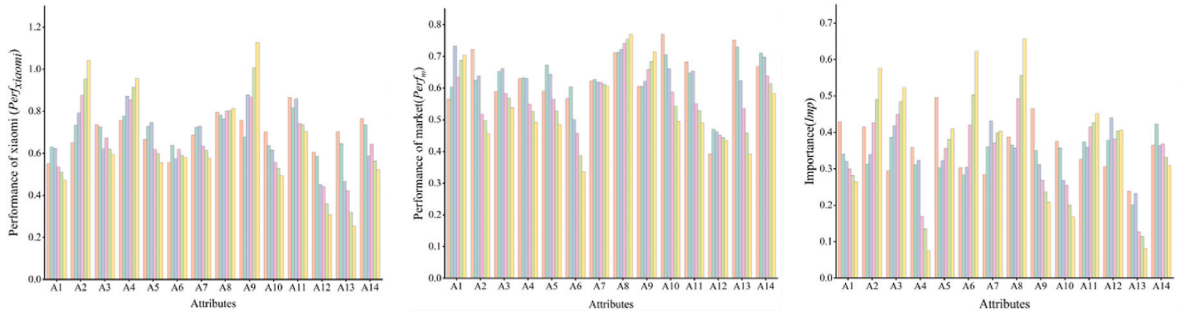



**Fig. 5.** Actual and forecasted results for all periods.

**Table 16**
Indicators for the D-SWOT matrix and factor identification results in actual periods.

| Attribute | Period3 | | | | Period4 | | | |
|---|---|---|---|---|---|---|---|---|
| | $ICP_{xiaomi}$ | $ICP_m$ | $GAP$ | Identified factor | $ICP_{xiaomi}$ | $ICP_m$ | $GAP$ | Identified factor |
| $A^1$ | −0.0611 | 0.0994 | −0.1095 | T | −0.1276 | 0.0572 | −0.0993 | T |
| $A^2$ | 0.1074 | 0.0048 | 0.1535 | S | 0.2129 | −0.0599 | 0.3584 | O |
| $A^3$ | −0.0636 | 0.0286 | −0.0413 | T | 0.0089 | 0.0052 | 0.0893 | S |
| $A^4$ | 0.1882 | −0.0030 | 0.2420 | O | 0.1911 | −0.0289 | 0.3056 | O |
| $A^5$ | 0.0621 | 0.0107 | 0.1023 | S | −0.0445 | −0.0131 | 0.0541 | O |
| $A^6$ | −0.1109 | −0.1327 | 0.0727 | O | −0.0449 | −0.1203 | 0.1610 | O |
| $A^7$ | 0.0451 | −0.0152 | 0.1112 | O | −0.0288 | 0.0396 | 0.0172 | O |
| $A^8$ | 0.0801 | 0.0884 | 0.0426 | S | 0.1387 | 0.1635 | 0.0608 | S |
| $A^9$ | 0.1937 | −0.0120 | 0.2565 | O | 0.2040 | 0.0814 | 0.2081 | S |
| $A^{10}$ | −0.0683 | 0.0274 | −0.0449 | T | −0.1063 | 0.0096 | −0.0304 | T |
| $A^{11}$ | 0.1744 | 0.0204 | 0.2050 | S | 0.0778 | −0.0273 | 0.1907 | O |
| $A^{12}$ | −0.2328 | −0.1710 | −0.0110 | W | −0.2215 | −0.1257 | −0.0102 | W |
| $A^{13}$ | −0.2180 | −0.0102 | −0.1569 | W | −0.2404 | −0.0421 | −0.1128 | W |
| $A^{14}$ | −0.0962 | 0.0643 | −0.1097 | T | −0.0196 | 0.0609 | 0.0051 | O |

**Table 17**
Indicators for the D-SWOT matrix and factor identification results in forecast periods.

| Attribute | Period5 | | | | Period6 | | | |
|---|---|---|---|---|---|---|---|---|
| | $ICP_{xiaomi}$ | $ICP_m$ | $GAP$ | Identified factor | $ICP_{xiaomi}$ | $ICP_m$ | $GAP$ | Identified factor |
| $A^1$ | −0.1411 | 0.1283 | −0.1781 | T | −0.1709 | 0.1683 | −0.2320 | T |
| $A^2$ | 0.3026 | −0.0618 | 0.4558 | O | 0.3994 | −0.0793 | 0.5859 | O |
| $A^3$ | −0.0314 | 0.0082 | 0.0518 | O | −0.0481 | 0.0025 | 0.0566 | O |
| $A^4$ | 0.2624 | −0.0327 | 0.3865 | O | 0.3131 | −0.0431 | 0.4634 | O |
| $A^5$ | −0.0530 | −0.0317 | 0.0701 | O | −0.0889 | −0.0507 | 0.0690 | O |
| $A^6$ | −0.0611 | −0.1717 | 0.2020 | O | −0.0629 | −0.1996 | 0.2440 | O |
| $A^7$ | −0.0377 | 0.0517 | 0.0021 | **O** | −0.0668 | 0.0707 | −0.0302 | **T** |
| $A^8$ | 0.1524 | 0.1949 | 0.0489 | S | 0.1709 | 0.2338 | 0.0443 | S |
| $A^9$ | 0.3560 | 0.1248 | 0.3226 | S | 0.4845 | 0.1785 | 0.4132 | S |
| $A^{10}$ | −0.1232 | −0.0168 | −0.0149 | W | −0.1490 | −0.0394 | −0.0024 | W |
| $A^{11}$ | 0.0851 | −0.0304 | 0.2069 | O | 0.0613 | −0.0451 | 0.2136 | O |
| $A^{12}$ | −0.2910 | −0.1157 | −0.0838 | W | −0.3339 | −0.1005 | −0.1262 | W |
| $A^{13}$ | −0.3326 | −0.1018 | −0.1394 | W | −0.3891 | −0.1435 | −0.1384 | W |
| $A^{14}$ | −0.0871 | 0.0547 | −0.0503 | T | −0.1196 | 0.0474 | −0.0597 | T |

performance. All models were assessed on the same dataset, utilizing mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) as key indicators of predictive performance. Typically, lower values of these metrics indicate superior predictive accuracy. The comparative results of model prediction accuracy are summarized in Tables 19–21.

The grey prediction model exhibits notable superiority in medium- and short-term data forecasting, achieving mean MAPE values of 0.0395 for $Perf_{Xiaomi}$, 0.0225 for $Perf_m$, and 0.0550 for $Imp$. Compared to other models, it demonstrates exceptional accuracy, highlighting its suitability for practical applications.

### 5.3. Comparison of SWOT analysis

The SWOT analysis method proposed in this study was compared with those of Phadermrod et al. (2019), Cheng et al. (2021), and Wu et al. (2024b). According to Phadermrod et al. (2019), an attribute $A^i$ is classified as a strength for a company if it is also a strength for its competitors. If $A^i$ is a strength for the company but a weakness for competitors, it is considered an opportunity. Conversely, if $A^i$ is a weakness for the company but a strength for competitors, it is categorized as a threat. If $A^i$ is a weakness for both, it remains classified as a weakness. According to Cheng et al. (2021), $A^i$ is considered a strength if it consistently receives positive feedback on new products or transitions from negative to positive reviews. Conversely, it is seen as a weakness if it continues to receive negative feedback or shifts from positive to

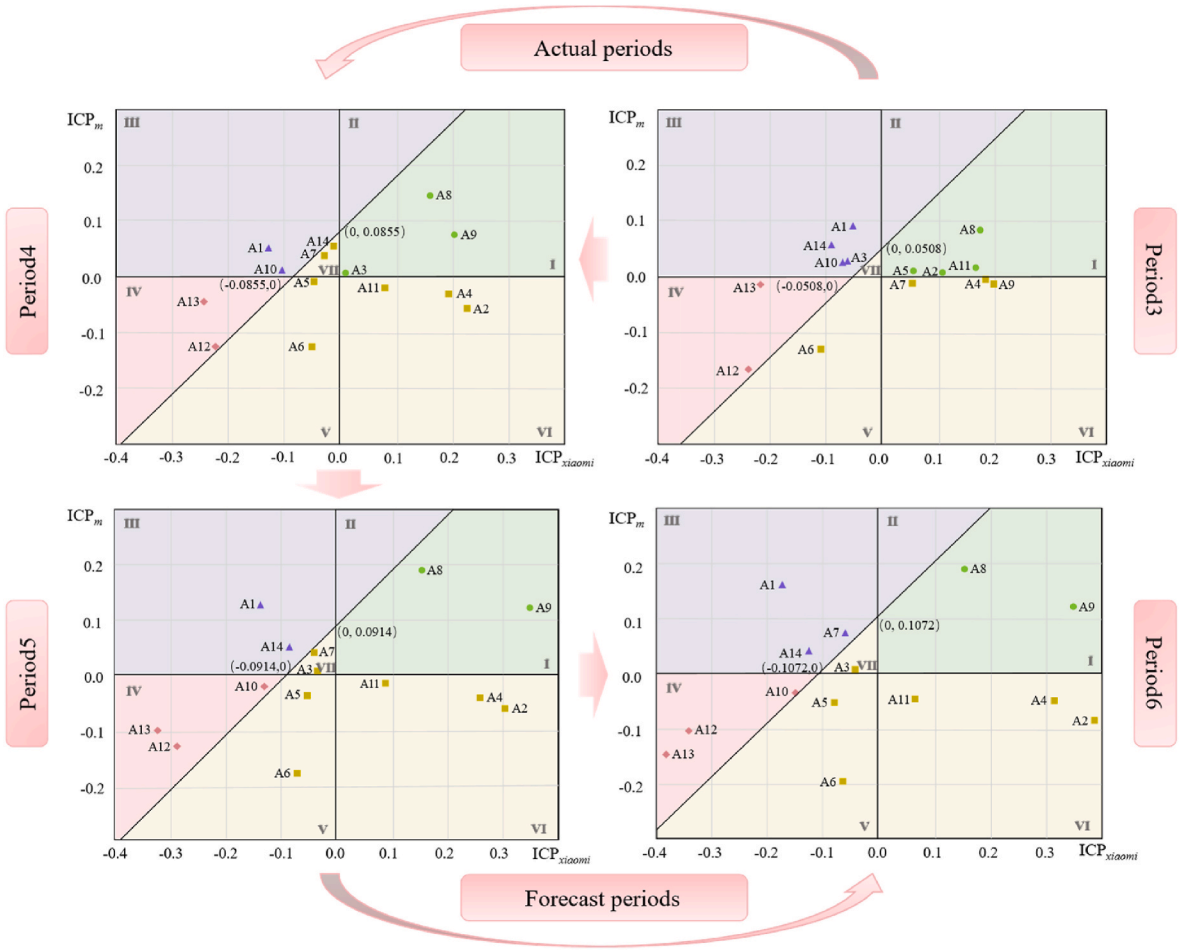



**Fig. 6.** The D-SWOT matrix for the actual and forecast periods.

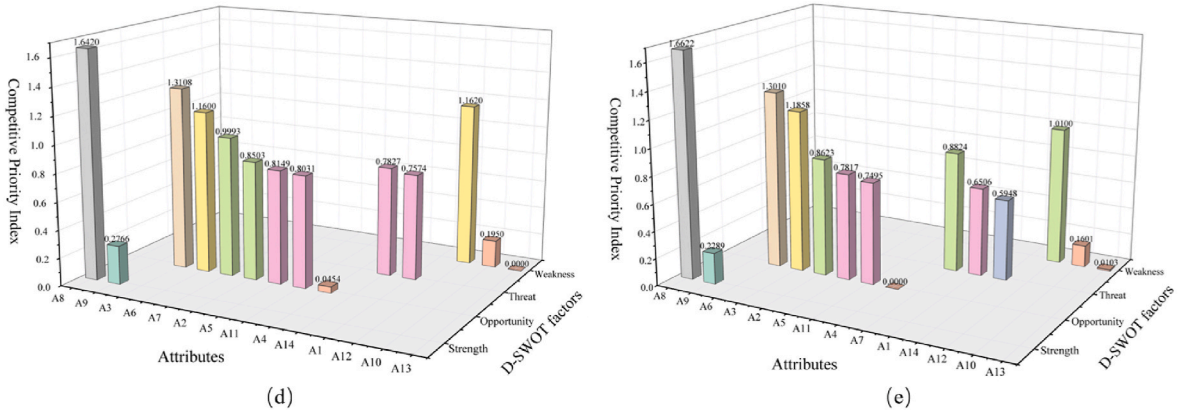



**Fig. 7.** Evolution of competitive priorities from period 5(d) to period 6(e).

**Table 18**
Priority changes of the key attributes.

| Attribute | Priority | Transformation |
|---|---|---|
| $A^3$ | Decrease | / |
| $A^6$ | Increase | / |
| $A^7$ | Decrease | O→T |
| $A^{14}$ | Decrease | / |

negative. It is regarded as an opportunity if $A^i$ performs well for the FP but poorly for competitors. On the other hand, if the attribute of FP underperforms while the competitor's corresponding attribute excels, it constitutes a threat. For instance, Table 22 compares our method with these two approaches using data from period 4.

However, the high proportion of positive evaluations in the mined UGC led to an overall positive attribute performance, making the method proposed by Cheng et al. (2021) less reliable. As a result, this approach classified all attributes in this study as strengths, which is inappropriate. The method by Phadermrod et al. (2019) often

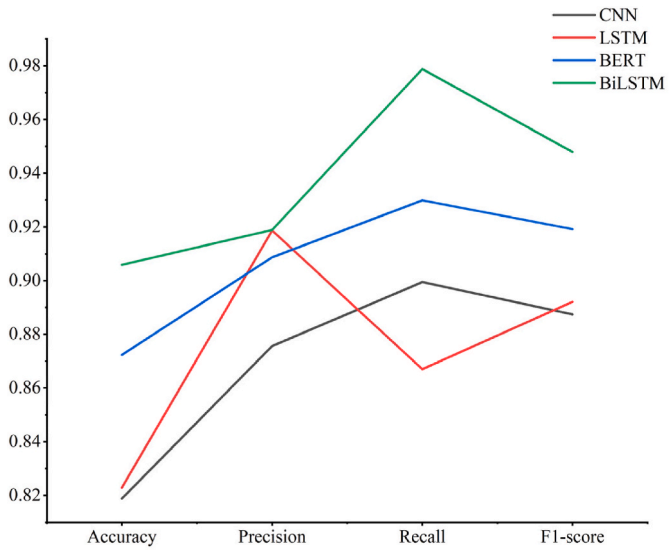



**Fig. 8.** Comparison results of sentiment analysis models.

misclassifies opportunities as threats or weaknesses. For instance, $A^5$ and $A^7$ illustrate this issue. Although $A^5$ is an internal weakness for both the Xiaomi Pad 6 Pro and the competitive market, it should be regarded as an opportunity because the Xiaomi Pad 6 Pro outperforms its competitors. Similarly, although the $A^7$ is an internal weakness for the Xiaomi Pad 6 Pro, it should also be considered an opportunity, given that it outperforms the competitive market.

Wu et al. (2024b) consider performance differences among products (Fig. 9). Comparative analysis reveals that, although various symbols are used to denote factor types, the method lacks a clear definition of the locations of strengths, weaknesses, opportunities, and threats, resulting in less intuitive visualizations. Therefore, this study introduces the $\eta$-translation line to explicitly define regional divisions while incorporating variations in $\eta$ values to track factor changes more effectively. Furthermore, this study incorporates time evolution trends into the SWOT matrix, enhancing its dynamic forecasting capability and allowing predictions of future evolution trends and foresight of factor changes based on actual data. For instance, in period 5 (see Fig. 6), the forecasts indicate that $A^3$ will shift from a strength to an opportunity, $A^{10}$ from a threat to a weakness, and $A^{14}$ from an opportunity to a threat. Quantitative forecasts from historical data enhance SWOT's foresight, improving the enterprise's ability to anticipate market changes. Thus, based on the above analysis, the D-SWOT method proposed in this study demonstrates greater rationality and foresight.

### 5.4. Comparison of priority indices

To illustrate the advantages of the CPI introduced in this study, Table 23 presents a comparison of this index with importance in period 4, using $A^5$ and $A^2$ as examples. Based on importance ranking, $A^2$ is positioned above $A^5$ and thus holds a higher priority when no additional factors are considered. However, when considering performance differences, $A^5$ exhibits a smaller gap. According to Equation (14), after standardization, the importance of $A^5$ exceeds its performance difference, leading to $CPI_t = 2Imp_t' - GAP_t'$, whereas $A^2$'s importance is lower than its performance difference, resulting in $CPI_t = Imp_t'$. As a result, $A^5$'s priority (0.9004) surpasses that of $A^2$ (0.8192). This approach prioritizes attributes with high importance and minimal differences, enhancing the precision of strategic decisions and optimizing resource allocation.

Consequently, compared to other methods that rely solely on the importance for resource allocation and factor improvement order (Sun et al., 2020; Wu et al., 2024b), the CPI proposed in this study

**Table 19**
Comparison of prediction accuracy for $Perf_{xiaomi}$.

| Attribute | Grey prediction model | | | Grey-markov model | | | ARIMA model | | | Fractional grey model (r = 0.3) | | | Fractional grey model (r = 0.5) | | | Fractional grey model (r = 0.8) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE |
| $A^1$ | 0.0187 | 0.0200 | 0.0313 | 0.0330 | 0.0430 | 0.0811 | 0.1830 | 0.2810 | 0.3264 | 0.3241 | 0.3932 | 0.5602 | 0.2652 | 0.3028 | 0.4567 | 0.1235 | 0.1338 | 0.2114 |
| $A^2$ | 0.0048 | 0.0049 | 0.0061 | 0.0310 | 0.0430 | 0.0784 | 0.1930 | 0.3280 | 0.2902 | 0.4650 | 0.5113 | 0.5654 | 0.3443 | 0.3799 | 0.4181 | 0.1416 | 0.1577 | 0.1715 |
| $A^3$ | 0.0344 | 0.0364 | 0.0524 | 0.0346 | 0.0442 | 0.0833 | 0.2170 | 0.3720 | 0.3016 | 0.3689 | 0.4425 | 0.5680 | 0.2799 | 0.3322 | 0.4289 | 0.1138 | 0.1405 | 0.1726 |
| $A^4$ | 0.0257 | 0.0273 | 0.0306 | 0.0336 | 0.0435 | 0.0810 | 0.2230 | 0.3810 | 0.2896 | 0.4910 | 0.5396 | 0.5765 | 0.3736 | 0.4065 | 0.4395 | 0.1656 | 0.1754 | 0.1956 |
| $A^5$ | 0.0330 | 0.0352 | 0.0470 | 0.0344 | 0.0441 | 0.0828 | 0.2180 | 0.3390 | 0.3265 | 0.3834 | 0.4628 | 0.5624 | 0.3145 | 0.3578 | 0.4601 | 0.1494 | 0.1612 | 0.2167 |
| $A^6$ | 0.0244 | 0.0258 | 0.0406 | 0.0335 | 0.0434 | 0.0821 | 0.1660 | 0.2810 | 0.2924 | 0.3278 | 0.3908 | 0.5478 | 0.2550 | 0.2953 | 0.4241 | 0.1039 | 0.1244 | 0.1715 |
| $A^7$ | 0.0230 | 0.0244 | 0.0328 | 0.0333 | 0.0433 | 0.0812 | 0.2060 | 0.3470 | 0.3016 | 0.3822 | 0.4603 | 0.5611 | 0.3110 | 0.3535 | 0.4556 | 0.1444 | 0.1559 | 0.2105 |
| $A^8$ | 0.0122 | 0.0129 | 0.0157 | 0.0322 | 0.0428 | 0.0794 | 0.2130 | 0.3980 | 0.2678 | 0.4357 | 0.4979 | 0.5556 | 0.3337 | 0.3759 | 0.4252 | 0.1395 | 0.1579 | 0.1773 |
| $A^9$ | 0.0492 | 0.0526 | 0.0608 | 0.0361 | 0.0459 | 0.0843 | 0.2660 | 0.3910 | 0.3441 | 0.5066 | 0.5312 | 0.6122 | 0.3740 | 0.3933 | 0.4515 | 0.1690 | 0.1743 | 0.2058 |
| $A^{10}$ | 0.0088 | 0.0094 | 0.0146 | 0.0318 | 0.0427 | 0.0793 | 0.2040 | 0.3530 | 0.2965 | 0.3453 | 0.4097 | 0.5881 | 0.2673 | 0.3100 | 0.4552 | 0.1217 | 0.1336 | 0.2064 |
| $A^{11}$ | 0.0362 | 0.0385 | 0.0445 | 0.0348 | 0.0444 | 0.0825 | 0.2500 | 0.4350 | 0.2937 | 0.4516 | 0.5347 | 0.5654 | 0.3647 | 0.4104 | 0.4567 | 0.1707 | 0.1830 | 0.2129 |
| $A^{12}$ | 0.0256 | 0.0269 | 0.0541 | 0.0336 | 0.0434 | 0.0835 | 0.1910 | 0.3100 | 0.3371 | 0.3695 | 0.3803 | 0.7844 | 0.2315 | 0.2707 | 0.5126 | 0.0921 | 0.1138 | 0.2054 |
| $A^{13}$ | 0.0251 | 0.0261 | 0.0521 | 0.0336 | 0.0434 | 0.0833 | 0.2370 | 0.3620 | 0.3736 | 0.4240 | 0.4250 | 0.8630 | 0.2814 | 0.3076 | 0.6100 | 0.1023 | 0.1276 | 0.2328 |
| $A^{14}$ | 0.0446 | 0.0473 | 0.0706 | 0.0357 | 0.0453 | 0.0853 | 0.2430 | 0.3910 | 0.3342 | 0.4102 | 0.4564 | 0.6534 | 0.2715 | 0.3351 | 0.4382 | 0.1101 | 0.1425 | 0.1747 |
| Average | 0.0261 | 0.0277 | 0.0395 | 0.0337 | 0.0437 | 0.0820 | 0.2150 | 0.3549 | 0.3125 | 0.4061 | 0.4597 | 0.6117 | 0.3048 | 0.3451 | 0.4594 | 0.1320 | 0.1487 | 0.1975 |

**Table 20**
Comparison of prediction accuracy for $Perf_m$.

| Attribute | Grey prediction model | | | Grey-markov model | | | ARIMA model | | | Fractional grey model (r = 0.3) | | | Fractional grey model (r = 0.5) | | | Fractional grey model (r = 0.8) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE |
| $A^1$ | 0.0507 | 0.0538 | 0.0757 | 0.0360 | 0.0460 | 0.0858 | 0.1940 | 0.2930 | 0.3264 | 0.3951 | 0.4354 | 0.5879 | 0.3077 | 0.3327 | 0.4594 | 0.1464 | 0.1541 | 0.2187 |
| $A^2$ | 0.0307 | 0.0327 | 0.0514 | 0.0340 | 0.0440 | 0.8323 | 0.2360 | 0.3670 | 0.3449 | 0.3502 | 0.4117 | 0.6051 | 0.2705 | 0.3146 | 0.4679 | 0.1303 | 0.1416 | 0.2228 |
| $A^3$ | 0.0202 | 0.0214 | 0.0316 | 0.0330 | 0.0431 | 0.0811 | 0.1860 | 0.2980 | 0.3121 | 0.3500 | 0.4152 | 0.5626 | 0.2821 | 0.3187 | 0.4530 | 0.1302 | 0.1403 | 0.2084 |
| $A^4$ | 0.0184 | 0.0196 | 0.0303 | 0.0328 | 0.0430 | 0.0810 | 0.1790 | 0.3170 | 0.2887 | 0.3290 | 0.4024 | 0.5591 | 0.2696 | 0.3084 | 0.4567 | 0.1252 | 0.1356 | 0.2109 |
| $A^5$ | 0.0123 | 0.0132 | 0.0196 | 0.0322 | 0.0428 | 0.0798 | 0.1870 | 0.3000 | 0.3141 | 0.3408 | 0.4147 | 0.5646 | 0.2764 | 0.3184 | 0.4560 | 0.1273 | 0.1390 | 0.2089 |
| $A^6$ | 0.0112 | 0.0117 | 0.0221 | 0.0321 | 0.0427 | 0.0801 | 0.1840 | 0.2890 | 0.3335 | 0.3382 | 0.3677 | 0.6873 | 0.2264 | 0.2753 | 0.4734 | 0.1011 | 0.1173 | 0.2099 |
| $A^7$ | 0.0017 | 0.0018 | 0.0028 | 0.0311 | 0.0425 | 0.0780 | 0.1600 | 0.3110 | 0.2564 | 0.3450 | 0.4002 | 0.5581 | 0.2690 | 0.3032 | 0.4349 | 0.1159 | 0.1284 | 0.1874 |
| $A^8$ | 0.0022 | 0.0023 | 0.0030 | 0.0311 | 0.0426 | 0.0781 | 0.1820 | 0.3560 | 0.2561 | 0.4095 | 0.4627 | 0.5607 | 0.3132 | 0.3492 | 0.4289 | 0.1332 | 0.1472 | 0.1824 |
| $A^9$ | 0.0046 | 0.0049 | 0.0074 | 0.0314 | 0.0426 | 0.0785 | 0.1600 | 0.3030 | 0.2633 | 0.3578 | 0.3980 | 0.5613 | 0.2695 | 0.2990 | 0.4225 | 0.1126 | 0.1253 | 0.1763 |
| $A^{10}$ | 0.0076 | 0.0082 | 0.0118 | 0.0317 | 0.0426 | 0.0790 | 0.2180 | 0.3870 | 0.2888 | 0.3952 | 0.4517 | 0.6288 | 0.2877 | 0.3399 | 0.4607 | 0.1322 | 0.1463 | 0.2103 |
| $A^{11}$ | 0.0253 | 0.0269 | 0.0407 | 0.0336 | 0.0434 | 0.0821 | 0.2040 | 0.3450 | 0.3079 | 0.3439 | 0.4166 | 0.5719 | 0.2781 | 0.3197 | 0.4616 | 0.1314 | 0.1423 | 0.2164 |
| $A^{12}$ | 0.0007 | 0.0007 | 0.0015 | 0.0309 | 0.0425 | 0.0779 | 0.1230 | 0.2000 | 0.3036 | 0.2558 | 0.2978 | 0.5590 | 0.2006 | 0.2263 | 0.4382 | 0.0875 | 0.0963 | 0.1910 |
| $A^{13}$ | 0.0012 | 0.0014 | 0.0019 | 0.0310 | 0.0425 | 0.0779 | 0.2170 | 0.3790 | 0.2956 | 0.4331 | 0.4612 | 0.7231 | 0.2853 | 0.3424 | 0.4925 | 0.1286 | 0.1466 | 0.2205 |
| $A^{14}$ | 0.0107 | 0.0114 | 0.0157 | 0.0320 | 0.0427 | 0.0794 | 0.1950 | 0.3350 | 0.2916 | 0.3733 | 0.4484 | 0.5590 | 0.3008 | 0.3425 | 0.4494 | 0.1359 | 0.1481 | 0.2026 |
| Average | 0.0141 | 0.0150 | 0.0225 | 0.0324 | 0.0431 | 0.1336 | 0.1875 | 0.3200 | 0.2988 | 0.3584 | 0.4131 | 0.5920 | 0.2741 | 0.3136 | 0.4539 | 0.1241 | 0.1363 | 0.2047 |



**Table 21**
Comparison of prediction accuracy for *Imp*.

| Attribute | Grey prediction model | | | Grey-markov model | | | ARIMA model | | | Fractional grey model (r = 0.3) | | | Fractional grey model (r = 0.5) | | | Fractional grey model (r = 0.8) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE | MAE | RMSE | MAPE |
| $A^1$ | 0.0006 | 0.0007 | 0.0018 | 0.0310 | 0.0430 | 0.0780 | 0.1375 | 0.2189 | 0.3416 | 0.2010 | 0.2267 | 0.6439 | 0.1412 | 0.1667 | 0.4552 | 0.0634 | 0.0706 | 0.2034 |
| $A^2$ | 0.0119 | 0.0123 | 0.0342 | 0.0321 | 0.0427 | 0.0814 | 0.1521 | 0.2163 | 0.3890 | 0.2069 | 0.2130 | 0.5729 | 0.1384 | 0.1487 | 0.3727 | 0.0516 | 0.0616 | 0.1344 |
| $A^3$ | 0.0005 | 0.0006 | 0.0011 | 0.0309 | 0.0425 | 0.0778 | 0.1065 | 0.1551 | 0.3335 | 0.2420 | 0.2697 | 0.5650 | 0.1819 | 0.2014 | 0.4247 | 0.0763 | 0.0841 | 0.1782 |
| $A^4$ | 0.0374 | 0.0398 | 0.1444 | 0.0351 | 0.0448 | 0.0948 | 0.1370 | 0.1932 | 0.4940 | 0.1560 | 0.1823 | 0.6164 | 0.1283 | 0.1605 | 0.5139 | 0.0691 | 0.0849 | 0.2528 |
| $A^5$ | 0.0030 | 0.0032 | 0.0093 | 0.0311 | 0.0426 | 0.0786 | 0.1866 | 0.2667 | 0.4537 | 0.1910 | 0.1941 | 0.5857 | 0.1321 | 0.1385 | 0.3970 | 0.0543 | 0.0594 | 0.1618 |
| $A^6$ | 0.0187 | 0.0192 | 0.0587 | 0.0329 | 0.0430 | 0.0840 | 0.1115 | 0.1592 | 0.3484 | 0.1926 | 0.2051 | 0.5563 | 0.1288 | 0.1452 | 0.3618 | 0.0446 | 0.0602 | 0.1169 |
| $A^7$ | 0.0293 | 0.0311 | 0.0742 | 0.0340 | 0.0437 | 0.0856 | 0.1193 | 0.1529 | 0.3771 | 0.2306 | 0.2569 | 0.5831 | 0.1807 | 0.1966 | 0.4585 | 0.0861 | 0.0909 | 0.2183 |
| $A^8$ | 0.0300 | 0.0314 | 0.0782 | 0.0341 | 0.0438 | 0.0861 | 0.1382 | 0.2047 | 0.3405 | 0.2222 | 0.2385 | 0.5313 | 0.1507 | 0.1733 | 0.3517 | 0.0508 | 0.0744 | 0.1095 |
| $A^9$ | 0.0023 | 0.0026 | 0.0075 | 0.0311 | 0.0426 | 0.0785 | 0.1580 | 0.2406 | 0.3748 | 0.2630 | 0.2659 | 0.8639 | 0.1514 | 0.1776 | 0.5193 | 0.0653 | 0.0743 | 0.2236 |
| $A^{10}$ | 0.0150 | 0.0158 | 0.0538 | 0.0325 | 0.0429 | 0.0835 | 0.1218 | 0.1925 | 0.3511 | 0.2346 | 0.2371 | 0.8336 | 0.1462 | 0.1664 | 0.5484 | 0.0562 | 0.0697 | 0.2153 |
| $A^{11}$ | 0.0156 | 0.0165 | 0.0415 | 0.0325 | 0.0429 | 0.0822 | 0.1057 | 0.1657 | 0.3135 | 0.2117 | 0.2397 | 0.5472 | 0.1576 | 0.1793 | 0.4057 | 0.0616 | 0.0748 | 0.1565 |
| $A^{12}$ | 0.0267 | 0.0283 | 0.0654 | 0.0337 | 0.0435 | 0.0847 | 0.1204 | 0.1613 | 0.3628 | 0.2356 | 0.2639 | 0.5793 | 0.1849 | 0.2019 | 0.4560 | 0.0874 | 0.0923 | 0.2152 |
| $A^{13}$ | 0.0305 | 0.0323 | 0.1642 | 0.0342 | 0.0439 | 0.0962 | 0.0903 | 0.1270 | 0.4606 | 0.1079 | 0.1314 | 0.5957 | 0.0883 | 0.1097 | 0.4848 | 0.0495 | 0.0599 | 0.2496 |
| $A^{14}$ | 0.0135 | 0.0143 | 0.0358 | 0.0323 | 0.0428 | 0.0815 | 0.1173 | 0.1855 | 0.3159 | 0.2129 | 0.2535 | 0.5793 | 0.1626 | 0.1919 | 0.4415 | 0.0690 | 0.0813 | 0.1863 |
| Average | 0.0168 | 0.0177 | 0.0550 | 0.0327 | 0.0432 | 0.0838 | 0.1287 | 0.1885 | 0.3755 | 0.2077 | 0.2270 | 0.6181 | 0.1481 | 0.1684 | 0.4422 | 0.0632 | 0.0742 | 0.1873 |

**Table 22**
Comparison results of SWOT analysis.

| Attribute | Cheng et al. (2021) | Phadermrod et al. (2019) | Our method |
|---|---|---|---|
| $A^1$ | S | T | T |
| $A^2$ | S | O | O |
| $A^3$ | S | S | S |
| $A^4$ | S | O | O |
| $A^5$ | S | **W** | **O** |
| $A^6$ | S | **W** | **O** |
| $A^7$ | S | **T** | **O** |
| $A^8$ | S | S | S |
| $A^9$ | S | S | S |
| $A^{10}$ | S | T | T |
| $A^{11}$ | S | O | O |
| $A^{12}$ | S | W | W |
| $A^{13}$ | S | W | W |
| $A^{14}$ | S | **T** | **O** |

significantly enhances the decision-making process. In practice, this method enables companies to more precisely identify critical factors and develop enhancements for attributes that contribute to competitive advantages. For instance, companies can leverage this model to adjust priorities for highly valued attributes yet exhibit minimal market differentiation, enhancing their appeal to target consumers.

## 6. Conclusions and implications

### 6.1. Conclusions

This study introduces a competitive analysis framework driven by multisource data, viewed from the perspective of multiple competitors. It improves traditional SWOT analysis methods by addressing their subjectivity, lack of dynamic perspectives, and disregard for factor importance, thus advancing the evolution of SWOT construction and analysis techniques. The first three steps of the framework lay the foundation for constructing the D-SWOT matrix, which includes competitor identification, attribute importance calculation, performance evaluation, and dynamic forecasting based on a grey prediction model. Ultimately, the D-SWOT matrix is constructed, and factor priorities are evaluated based on the outcomes.

The Xiaomi Pad 6 Pro case analysis reveals that in period 5, $A^3$ shifted from a strength to an opportunity, $A^{10}$ changed from a threat to a weakness, and $A^{14}$ transitioned from an opportunity to a threat. In period 6, $A^7$ changed from an opportunity to a threat. These changes also reflect shifts in the competitive market level. For example, $A^3$'s transition from a strength to an opportunity is due to Xiaomi Pad 6 Pro's performance in this attribute becoming an internal weakness, yet still outperforming the competition. Thus, this can be viewed as a relative improvement in Xiaomi Pad 6 Pro's IAP or a relative decrease in the competitive market's IAP.

We assessed the key attributes and their evolution in each period

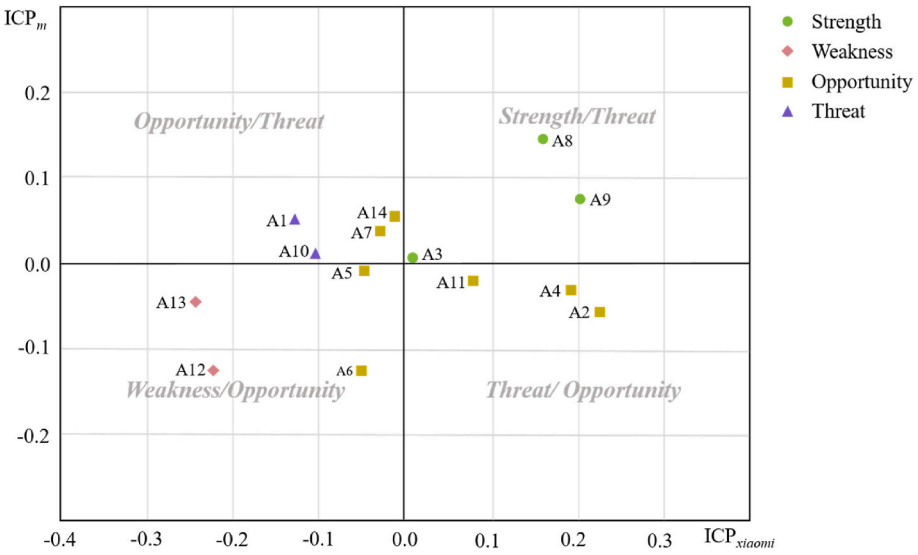



**Fig. 9.** SWOT matrix proposed by Wu et al. (2024b).

**Table 23**
Comparison of *Imp* and *CPI*.

| Attribute | *Imp* | *GAP* | *Imp'* | *GAP'* | *CPI* | Identified factor | Rank |
|---|---|---|---|---|---|---|---|
| $A^8$ | 0.4920 | 0.0608 | 1.0000 | 0.3684 | 1.6316 | S | 1 |
| $A^3$ | 0.4490 | 0.0893 | 0.8822 | 0.4289 | 1.3355 | S | 2 |
| $A^9$ | 0.2690 | 0.2081 | 0.3890 | 0.6812 | 0.3890 | S | 3 |
| $A^{14}$ | 0.3680 | 0.0051 | 0.6603 | 0.2502 | 1.0703 | O | 1 |
| $A^7$ | 0.3710 | 0.0172 | 0.6685 | 0.2760 | 1.0610 | O | 2 |
| $A^6$ | 0.4200 | 0.1610 | 0.8027 | 0.5811 | 1.0244 | O | 3 |
| $A^{11}$ | 0.4140 | 0.1907 | 0.7863 | 0.6441 | 0.9285 | O | 4 |
| $A^5$ | 0.3560 | 0.0541 | 0.6274 | 0.3544 | 0.9004 | O | 5 |
| $A^2$ | 0.4260 | 0.3584 | 0.8192 | 1.0000 | 0.8192 | O | 6 |
| $A^4$ | 0.1700 | 0.3056 | 0.1178 | 0.8880 | 0.1178 | O | 7 |
| $A^1$ | 0.2990 | −0.0993 | 0.4712 | 0.0288 | 0.9137 | T | 1 |
| $A^{10}$ | 0.2550 | −0.0304 | 0.3507 | 0.1750 | 0.5263 | T | 2 |
| $A^{12}$ | 0.3820 | −0.0102 | 0.6986 | 0.2177 | 1.1795 | W | 1 |
| $A^{13}$ | 0.1270 | −0.1128 | 0.0000 | 0.0000 | 0.0000 | W | 2 |

regarding factor prioritization. In period 5, $A^8$ was the key strength, $A^3$ was the key opportunity, $A^{14}$ was the key threat, and $A^{12}$ was the key weakness. In period 6, $A^6$ became the key opportunity, the priority of $A^{14}$ declined, and $A^7$ shifted to a threat and became a key threat. These changes in priorities reflect the dynamic shifts in the market environment and user needs, urging companies to adjust their strategies promptly to address new challenges and seize potential opportunities.

### 6.2. Methodological implications

The methodological contributions of this study are as follows:

(1) This study presents a method for evaluating the competitive position of an FP from a multi-competitor perspective. Unlike traditional SWOT studies that compare the focal product against single or multiple competitors (Phadermrod et al., 2019; Srinivas and Rajendran, 2019), this approach innovatively treats the focal product and all its competitors as a single competitive market, accurately positioning the product in its specific competitive context. Consequently, by leveraging collected multisource data, this paper broadens the scope of competitive products, establishes a competitive market centered on the focal product, and offers a practical methodology for businesses to understand their products' competitive standings.
(2) This research innovatively incorporates the grey prediction model into SWOT analysis, diverging from the static analytical methods prevalent in prior studies (Lee et al., 2021; Wang et al., 2022). Offering objective forecasts based on existing data sequences reduces the subjectivity inherent in traditional SWOT analysis to a certain degree. Furthermore, forecasting the future states of attributes provides data-based dynamic inputs that transform SWOT from a static evaluation to a dynamic strategic tool adaptable to future changes. The case study also demonstrates the model's applicability to unstructured data, providing practical solutions for complex, incomplete data issues and advancing the development and innovation of prediction models in practical applications.
(3) This research advances the construction and analytical methodologies of SWOT analysis. While some scholars have suggested identifying SWOT factors through UGC data (Cheng et al., 2021; Lin et al., 2023), they have overlooked performance differences. Although Wu et al. (2024b) considered performance differences, they did not refine the definition of SWOT factor regions. In contrast, this study introduces the $\eta$-translation line. It proposes three D-SWOT matrices under varying $\eta$ values, defining the criteria for segmenting SWOT factors based on UGC data. This approach provides a novel visualization technique for tracking attribute evolution, offering a more rigorous foundation for decision-making in complex competitive environments.
(4) Utilizing performance differences, this study introduces a CPI to address the issue of factor improvement priorities in a multi-competitor environment. Unlike prior studies that solely relied on importance assessment results to dictate the sequence of improvements (Büyüközkan et al., 2021; Phadermrod et al., 2019; Wu et al., 2024b), the proposed CPI takes into account both performance differences and importance enabling accurate identification of attributes that possess strategic advantages, significantly boosting the product's market competitiveness. Overall, this research demonstrates that improvements in methodologies based on a multi-competitor perspective maintain the applicability and effectiveness of SWOT principles in the big data era.

### 6.3. Managerial implications

The approach proposed in this study offers businesses insights into product improvement and competitive strategy adjustments within a multi-competitor environment. The management implications, based on the tablet application case, are as follows:

(1) This study's findings allow business managers to monitor the FP's competitive positioning in the market. When applied to the FP, the proposed method reveals a set of related competitive submarkets, allowing managers of specific products to quickly assess the FP's market position relative to any competitor.
(2) This study integrates time evolution trends into the multi-competitor perspective, using the grey prediction model to forecast the required indicators, including attribute performance and attribute importance. By predicting these variables, we find that past data provides guiding significance for future development. By anticipating these changes, businesses can effectively seize potential market opportunities and prevent possible risks. Applying the grey prediction model has significant strategic value in business management decisions, providing robust data-driven support for companies to formulate proactive strategies in a complex and dynamic market environment.
(3) This study presents specific competitive analysis results and improvement strategies. The proposed D-SWOT matrix clearly illustrates the evolution of the FP's competitive position. For instance, $A^7$ in the case study is an opportunity attribute in period 5, but it transitions into a key threat attribute in period 6. If these changes can be foreseen, businesses can take proactive measures to prevent the threats posed by a decline in competitive position. Additionally, the proposed CPI emphasizes the importance of attributes with high significance but low differentiation. It helps businesses identify the attributes within each SWOT factor that should be prioritized for action. By proactively strengthening attributes that already possess a competitive advantage or improving those that perform poorly in comparison, the FP can gain a more advantageous competitive position.

### 6.4. Limitations and future research

Although the method proposed in this study offers methodological support for competitive analysis from a multi-competitor perspective, it still has some limitations. First, this study identifies competitors based on product name co-occurrence but does not further categorize competitor types. Future research could focus on competitive analysis of the FP across multiple competitive sub-markets to explore product competitiveness from various perspectives. Secondly, although this study employs multisource data, it does not consider internal company data. Future research could incorporate internal company data and apply the methods proposed in this study for competitive analysis. In addition, this study uses the LDA method for attribute extraction. Although the extracted attributes are good in interpretability, they have inherent limitations. In the future, it may be worth considering using integrated topic models and machine learning algorithms (such as Bertopic) for opinion mining of UGC.

## CRediT authorship contribution statement

**Yanlai Li:** Writing – review & editing, Supervision, Funding acquisition. **Huiru Yu:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation. **Zifan Shen:** Writing – review & editing, Supervision, Methodology.

## Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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